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Chemistry
Class 12

Sindh Textbook Board

(English Medium)

Based on the Sindh Textbook Board Book

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Chemistry of Representative Elements

Unit 2
Chemistry of Outer Transition (d-Block) Elements

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Chemistry of Representative Elements

UNIT 1

General Group Trends of Representative Elements

LESSON 1

The representative elements are the s-block and p-block elements of groups IA to VIIIA. Their properties change in regular trends down a group and across a period, and these trends can be explained from atomic size, shielding and effective nuclear charge.

1. Atomic Radii

Atomic radii of the alkali metals increase regularly from Li to Cs, because a new electron shell is added at each step and shielding increases. A famous exception in group IIIA is gallium: gallium has a smaller atomic radius than aluminium although it lies below it, because gallium comes after the first transition series and its ten 3d electrons shield the nuclear charge poorly. The increased effective nuclear charge pulls the outer electrons closer (d-block contraction), so gallium is slightly smaller than aluminium.

2. Ionization Energy

Ionization energy decreases from top to bottom in the s-block elements. Down the group a new shell is added at each step, so the valence electron is farther from the nucleus and is better shielded by the inner electrons. The effective nuclear attraction on it therefore decreases, and less energy is needed to remove it. In group IIIA the trend is irregular: ionization energy falls from B to Al, then rises slightly at Ga, falls at In and rises again at Tl. The rises at Ga and Tl are due to the poor shielding of the filled d and f electrons (d-block and f-block contractions), which increases the effective nuclear charge. The overall trend is a decrease with irregular increases at gallium and thallium.

3. Electronegativity

Electronegativities of the alkali metals decrease from Li to Cs (Li 1.0 to Cs 0.7). Down the group the atomic size increases and the valence electron is held less tightly because of greater distance and shielding, so the atom attracts a shared pair of electrons less strongly. As a result, metallic and electropositive character, and chemical reactivity, increase down the group.

4. Melting and Boiling Points

Melting and boiling points regularly decrease down group IA: in the alkali metals the metallic bonding weakens regularly as atomic size increases down the group, so the points decrease steadily from Li to Cs. In the other groups the trend is irregular because of changes in structure and bonding type. Boiling points of the halogens, on the other hand, increase down the group: the halogen molecules become larger and their electron clouds more easily polarised, so the van der Waals forces between the molecules become stronger and more heat is needed to separate them.

5. Electrical Conductivity

Alkali metals are good conductors of electricity. They have one loosely held valence electron per atom in a sea of delocalised electrons within the metallic lattice. These free electrons move readily under an applied potential, so the metals conduct electricity and heat very well.

Reactions of Representative Elements

LESSON 2

The representative elements show their characteristic chemistry in reactions with air, water, nitrogen and acids. The balanced equations below are the standard examples from the book.

1. Reaction with Air and Nitrogen

Sodium in excess air: sodium burns in excess air to form mainly sodium peroxide, a pale yellow solid: 2Na + O2 -> Na2O2. With a limited supply of air the normal oxide is formed instead: 4Na + O2 -> 2Na2O. Potassium is still more reactive and forms the superoxide KO2 on burning in excess air, while lithium forms only the normal oxide Li2O.

Magnesium with nitrogen: hot magnesium combines directly with nitrogen to form magnesium nitride, a greenish-yellow solid: 3Mg + N2 -> Mg3N2. The nitride is hydrolysed by warm water to give ammonia: Mg3N2 + 6H2O -> 3Mg(OH)2 + 2NH3. The general formula of the nitrides of the alkaline earth metals is M3N2, because the metal forms M2+ ions and nitrogen forms N3- ions, so three M2+ ions balance two N3- ions, as in Mg3N2 and Ca3N2.

2. Reactions with Alcohols and Alkalis

Potassium with ethyl alcohol: potassium reacts with ethanol to form potassium ethoxide with the evolution of hydrogen gas, more vigorously than sodium does: 2K + 2C2H5OH -> 2C2H5OK + H2. This shows the weakly acidic character of the -OH group of alcohols.

Ferric chloride with caustic soda: a brown precipitate of ferric hydroxide is formed in this double decomposition reaction: FeCl3 + 3NaOH -> Fe(OH)3 + 3NaCl. The reddish-brown Fe(OH)3 precipitate is a characteristic test for Fe3+ ions.

3. Reaction of Aluminium with Sulphuric Acid

Aluminium is oxidised by hot concentrated sulphuric acid, which itself is reduced to sulphur dioxide: 2Al + 6H2SO4 -> Al2(SO4)3 + 3SO2 + 6H2O. This shows that concentrated H2SO4 acts as a strong oxidising agent towards metals.

Flame Test of s-Block Elements

LESSON 3

1. The Flame Test

In a flame test a small amount of a metal salt on a clean platinum or nichrome wire is held in a Bunsen flame, and the metal imparts a characteristic colour to the flame. The heat of the flame excites the metal atoms, and they emit light of a characteristic colour on returning to the ground state. The test is used to identify alkali metal ions in a sample.

2. Flame Colours of Alkali Metals

- **Lithium:** crimson red
- **Sodium:** golden yellow
- **Potassium:** lilac
- **Rubidium:** red-violet
- **Caesium:** blue-violet

Sodium is the alkali metal that gives the yellow colour in a flame test; the golden yellow flame is so intense that it can mask the colours of other metals present.

Chemistry of Important Compounds of s-Block Elements

LESSON 4

1. Caustic Soda by the Castner-Kellner Cell

Caustic soda (NaOH) is obtained by the electrolysis of aqueous sodium chloride in the Castner-Kellner cell. The cell has a flowing mercury cathode at the bottom and titanium anode blocks in brine compartments separated by partitions, with a central compartment containing water or dilute NaOH. At the anode, chloride ions are discharged: 2Cl- -> Cl2 + 2e-, and chlorine gas is collected. At the mercury cathode, sodium ions are discharged into the mercury: Na+ + e- -> Na (amalgam). The sodium amalgam flows to the central compartment where it reacts with water: 2Na(Hg) + 2H2O -> 2NaOH + H2 + Hg, and the mercury is recycled, so pure caustic soda free from NaCl is obtained. The cathode in the Castner-Kellner cell is mercury.

2. Bleaching Powder

Bleaching powder is calcium oxychloride, Ca(OCl)Cl, a yellowish-white powder used as a bleaching and disinfecting agent. It is prepared by passing chlorine gas over dry slaked lime at about 40 C: Ca(OH)2 + Cl2 -> Ca(OCl)Cl + H2O. With water it gives nascent oxygen which does the bleaching; in the presence of moisture and CO2 it slowly releases chlorine, and with dilute hydrochloric acid it liberates chlorine rapidly: Ca(OCl)Cl + 2HCl -> CaCl2 + Cl2 + H2O.

3. Potassium Nitrate and Fireworks

The chemical commonly used in fireworks is potassium nitrate (saltpetre). It is the oxidising agent in gunpowder-type firework mixtures and supplies oxygen for rapid burning. Metal salts present with it produce the characteristic colours of fireworks.

Reactions of p-Block Elements

LESSON 5

The p-block elements include the non-metals and metalloids of groups IIIA to VIIIA. Their reactions below show the oxidising power of their compounds and the special behaviour of elements such as gallium.

1. Gallium and the d-Block Contraction

Gallium has a smaller atomic radius than aluminium despite being below aluminium in group IIIA. Gallium comes after the first transition series, and its ten 3d electrons shield the nuclear charge poorly. The increased effective nuclear charge pulls the outer electrons closer, a contraction called the d-block contraction, so gallium is slightly smaller than aluminium. For the same reason the ionization energy of group IIIA rises slightly at gallium instead of falling regularly.

2. Aluminium with Concentrated Sulphuric Acid

When a piece of aluminium is dipped into hot concentrated sulphuric acid, the acid oxidises the metal and is itself reduced to sulphur dioxide: 2Al + 6H2SO4 -> Al2(SO4)3 + 3SO2 + 6H2O. Concentrated sulphuric acid therefore acts as a strong oxidising agent towards p-block metals.

3. Double Decomposition: Ferric Chloride and Caustic Soda

When ferric chloride is mixed with an aqueous solution of caustic soda, a reddish-brown precipitate of ferric hydroxide separates: FeCl3 + 3NaOH -> Fe(OH)3 + 3NaCl. This precipitate is a characteristic test for Fe3+ ions.

Chemical Behaviour of Halogens

LESSON 6

1. Trends in the Halogen Group

Boiling point of the halogens increases down the group. The molecules become larger and their electron clouds more easily polarised, so the van der Waals forces between them become stronger, more heat is needed to separate them, and boiling point increases from F2 to I2.

2. Acidity of the Hydrogen Halides

The acidity of the hydrogen halides increases from HF to HI. Down the group the H-X bond becomes longer and weaker as the halogen atom becomes larger, so the bond breaks more easily to release H+ ions. Bond strength, and not electronegativity, controls the acidity here, so the order is HI > HBr > HCl > HF.

3. Fluorine: the Strongest Oxidising Agent

Fluorine is the smallest and most electronegative halogen, the F-F bond is weak and the hydration energy of the small F- ion is very high. It therefore gains electrons most readily and oxidises other substances most powerfully, making F2 the strongest oxidising agent. Oxidising power decreases down the group in the order F2 > Cl2 > Br2 > I2.

4. Disproportionation of Chlorine

In some reactions chlorine is oxidised and reduced at the same time, a process called disproportionation (auto oxidation-reduction), so it acts as its own oxidising and reducing agent. With cold dilute NaOH it gives chloride and hypochlorite: Cl2 + 2NaOH -> NaCl + NaOCl + H2O, in which chlorine goes from 0 to -I and to +I. With hot concentrated NaOH it forms chloride and chlorate: 3Cl2 + 6NaOH -> 5NaCl + NaClO3 + 3H2O.

Chemistry of Sulphuric Acid

LESSON 7

Sulphuric acid is called the king of chemicals; its old name is oil of vitriol, because concentrated sulphuric acid is an oily, dense liquid obtained historically from vitriol minerals. It is manufactured by the Contact process.

1. The Contact Process

The flow of the process is: sulphur (or iron pyrites) burner -> SO2 gas -> purification unit -> converter containing V2O5 at about 450 C -> SO3 gas -> absorption tower with concentrated H2SO4 -> oleum -> dilution with water -> concentrated H2SO4.

- **Step 1, burning:** S + O2 -> SO2
- **Step 2, catalytic oxidation:** 2SO2 + O2 -> 2SO3 in the presence of the catalyst vanadium pentoxide, V2O5, at about 450 C. This is the key step of the process.
- **Step 3, absorption:** SO3 + H2SO4 -> H2S2O7 (oleum), followed by H2S2O7 + H2O -> 2H2SO4. SO3 is not absorbed directly in water, because that produces a dense mist which is difficult to condense.

Diagonal Relationship of Representative Elements

LESSON 8

1. What is a Diagonal Relationship?

A diagonal relationship is the similarity in properties between an element and the element lying diagonally below it in the next group, caused by a balance of increasing size down the group and increasing charge across the period. Three such pairs among the representative elements are: lithium and magnesium, beryllium and aluminium, and boron and silicon.

2. Beryllium and Aluminium

The diagonal member of beryllium is aluminium. Beryllium lies diagonally next to aluminium in the periodic table and resembles it closely: both form amphoteric oxides and hydroxides that dissolve in acids as well as alkalis, and both form largely covalent chlorides. Beryllium also shows unique behaviour within its own group IIA: its oxide and hydroxide are amphoteric, unlike the basic oxides of the others; its compounds are largely covalent and it never shows a coordination number greater than four; and it does not react with water even on heating, whereas magnesium and the heavier metals react readily.

Chemistry of Outer Transition (d-Block) Elements

UNIT 2

General Features of Outer Transition Elements

LESSON 9

The d-block elements are called transition elements because their properties are intermediate between those of the highly reactive s-block metals and the p-block elements, and they form a bridge between them in the periodic table. They are called outer transition elements because their differentiating electron enters the outer (n-1)d subshell, in contrast to the inner transition (f-block) elements. Their atoms or ions have partially filled d orbitals, which gives them their characteristic properties. The 5d series lies in the sixth period and runs from lanthanum (La) to mercury (Hg); Sc to Zn is the 3d series, Y to Cd the 4d series and Ac to Cn the 6d series.

1. Variable Oxidation States

Transition elements show variable oxidation states because the (n-1)d and ns orbitals have very similar energies, so a variable number of electrons from both can take part in bonding. In the 3d series the common state is +2 (loss of 4s electrons), and the maximum oxidation state rises from Sc (+3) to Mn (+7), equal to the total number of 4s and 3d electrons. After manganese the maximum falls (Fe +6, Co +5, Ni +4, Cu +2, Zn +2) because the d electrons become held more tightly. Chromium (3d5 4s1) can use all six valence electrons in bonding, so its highest oxidation state is +6, shown in compounds such as K2Cr2O7 and CrO3, where chromium is a powerful oxidising agent.

2. Special Electronic Configurations

Chromium exists in the 4s1 3d5 configuration and not in 4s2 3d4, because a half-filled d subshell (3d5) has extra stability due to symmetrical electron distribution and maximum exchange energy. One electron therefore shifts from 4s to 3d to make the d subshell exactly half-filled, which makes 4s1 3d5 lower in energy and more stable.

3. Paramagnetic Behaviour

Paramagnetism is caused by unpaired electrons, and its strength increases with their number. Across the 3d series the number of unpaired electrons rises from Sc (1) to a maximum in the Mn/Fe region (Mn2+ and Fe3+ have 5) and then falls as pairing begins, reaching zero at Zn (3d10). Hence paramagnetism first increases, becomes maximum in the middle of the series, and then decreases; Zn2+ and Sc3+ are diamagnetic. Fe3+ (3d5) is the most strongly paramagnetic of the common ions Fe2+, Fe3+, Co2+ and Cr3+.

4. Colour Formation

Most transition metal compounds are coloured because their ions have partially filled d orbitals which split into two sets (t2g and eg) in the presence of ligands. Electrons absorb visible light of a definite colour to jump between these sets (d-d transition), and the compound shows the complementary colour. Cu2+ (3d9) has one unpaired electron that can undergo d-d transition, so its compounds are blue. Zn2+ has the configuration 3d10 with all electrons paired, so no d-d transition is possible; it absorbs no visible light and its compounds are white or colourless. Ions with d0 (Sc3+, Ti4+) or d10 (Zn2+, Cu+) configurations are likewise colourless.

5. Melting Points and Binding Energy

Binding energy (cohesive energy) is the energy required to break one mole of a metal into isolated gaseous atoms; it measures the strength of metallic bonding. Melting point and binding energy depend on the number of unpaired d electrons taking part in bonding: both increase from Sc to a maximum around the middle of the series (Cr, Mo, W group; binding energy maximum around V-Cr-Mn) and then decrease as electron pairing begins. Binding energy is least for zinc, whose completely filled 3d10 configuration contributes no unpaired electrons to metallic bonding, so its melting point and hardness are also the lowest in the 3d series.

6. Alloy Formation

Transition elements form alloys readily because their atomic sizes are very similar, differing only slightly across a series, so atoms of one metal can easily replace atoms of another in the lattice without distorting it much. They therefore mix readily in the molten state and form solid solutions called alloys, such as brass, bronze and stainless steel.

7. Catalytic Properties

Transition elements and their compounds act as catalysts in many reactions because they can show several oxidation states, accept and donate electrons easily, and provide a suitable surface on which reactants are adsorbed. Reactants form temporary bonds with the metal surface or the metal ion, which lowers the activation energy. Examples are Fe in the Haber process for ammonia, V2O5 in the Contact process, Ni in the hydrogenation of oils and Pt in catalytic converters.

8. Electroplating

Zinc, tin and chromium are commonly deposited by electroplating to protect or decorate metal surfaces. Manganese is too reactive and difficult to deposit from aqueous solution, so it is not used for electroplating. When manganese is oxidised in air it forms the mixed oxide Mn3O4 (a mixture of MnO and Mn2O3): 3Mn + 2O2 -> Mn3O4.

Nomenclature of Coordination Compounds

LESSON 10

1. Coordination Complexes and Ligands

A coordination complex is a compound in which a central metal atom or ion is bonded to a fixed number of surrounding ions or molecules (ligands) by coordinate bonds, for example [Cu(NH3)4]2+. The number of coordinate bonds formed by the ligands with the metal is the coordination number: in Na4[Co(C2O4)3] the coordination number of cobalt is 6, because oxalate is a bidentate ligand donating two electron pairs per ligand, and three oxalate ligands form six coordinate bonds.

A chelating ligand is a polydentate ligand that binds through two or more donor atoms at the same time and forms a ring structure with the metal; the resulting complex is called a chelate, for example oxalate and EDTA complexes. Ligands are classified by the number of donor atoms: monodentate ligands donate through one atom (NH3, Cl-, CN-), bidentate ligands through two atoms (oxalate C2O4 2-, ethylenediamine), and polydentate ligands through many atoms (EDTA is hexadentate). Ligands may also be neutral (NH3, H2O, CO) or negatively charged (Cl-, CN-, C2O4 2-). Oxalate, C2O4 2-, is a bidentate ligand because it has two oxygen donor atoms through which it binds at two positions; OH-, Cl- and CN- are monodentate.

2. IUPAC Names of Complexes

- **Na2[Pt(OH)4]:** sodium tetrahydroxoplatinate(II). The complex anion contains Pt with four hydroxo ligands; charge balance with two Na+ ions shows Pt is in the +2 oxidation state. The name of an anionic complex ends in -ate and the oxidation state is shown in Roman numerals.
- **K2[Fe(CN)5NO]:** potassium pentacyanonitrosylferrate(II). This is the nitroprusside ion, containing five cyano ligands and one nitrosyl (NO) ligand around iron(II). The ligands are named alphabetically and the anionic complex of iron takes the name ferrate.
- **[Zn(NH3)4]2+:** tetraamminezinc(II) ion. The cation contains four ammine (NH3) ligands around zinc(II); the complex is cationic, so the metal keeps its ordinary name. Zinc shows only the +2 oxidation state in its complexes.
- **[Ni(SCN)4]2-:** tetrathiocyanatonickelate(II) ion. The anion contains four thiocyanato ligands around nickel(II), and an anionic complex of nickel is named as a nickelate.

Chemistry of Some Important Transition Elements

LESSON 11

1. Important Reactions

- **Copper with concentrated nitric acid:** Cu + 4HNO3 -> Cu(NO3)2 + 2NO2 + 2H2O. The acid oxidises copper and is itself reduced to nitrogen dioxide; brown fumes of NO2 are evolved and a blue solution of copper nitrate is formed.
- **Copper with hot concentrated sulphuric acid:** Cu + 2H2SO4 -> CuSO4 + SO2 + 2H2O. Copper does not react with dilute sulphuric acid because it lies below hydrogen in the reactivity series.
- **Permanganate with oxalic acid:** 2KMnO4 + 5H2C2O4 + 3H2SO4 -> K2SO4 + 2MnSO4 + 10CO2 + 8H2O. The purple colour of KMnO4 is discharged, which is

used in volumetric analysis.

- **Dichromate with ferrous sulphate:** $K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$. The orange colour of dichromate changes to green.
- **Manganese with dilute sulphuric acid:** $Mn + H_2SO_4 \rightarrow MnSO_4 + H_2$. Manganese lies above hydrogen in the reactivity series; a pale pink solution of manganese sulphate is obtained.
- **Iron with chlorine:** $2Fe + 3Cl_2 \rightarrow 2FeCl_3$. Chlorine is a strong oxidising agent and takes iron to its higher oxidation state; brown fumes of $FeCl_3$ are produced.

2. Chromate–Dichromate Equilibrium and pH

In aqueous solution chromate and dichromate exist in equilibrium: $2CrO_4^{2-}$ (yellow) $+ 2H^+ \leftrightarrow Cr_2O_7^{2-}$ (orange) $+ H_2O$. Adding acid (lowering pH) shifts the equilibrium to the right and the solution turns orange due to dichromate. Adding alkali (raising pH) removes H^+ ions, shifts the equilibrium to the left, and the solution becomes yellow due to chromate.

3. Extraction and Refining of Copper

Chalcopyrite, $CuFeS_2$, is first concentrated by froth flotation, the step in which the gangue (earthy impurities) is eliminated. The concentrate is roasted in a reverberatory furnace: $2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$, and the roasted ore is smelted with silica so that FeO is removed as slag ($FeO + SiO_2 \rightarrow FeSiO_3$), leaving copper matte ($Cu_2S + FeS$). In the Bessemer converter, air is blown through the matte: $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$ and $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$, giving blister copper of about 98% purity. Finally it is refined electrolytically using thick plates of blister copper as anode and thin sheets of pure copper as cathode in acidified $CuSO_4$ solution; pure copper (99.99%) deposits on the cathode while impurities collect as anode mud.

4. Important Alloys

- **Stainless steel:** about 74% iron, 18% chromium and 8% nickel. It resists rusting, does not react with food and is easy to clean, so it is used for cutlery, surgical instruments, fry pans and chemical plant.
- **Brass:** about 70% copper and 30% zinc; used for utensils, taps, locks and musical instruments.
- **Bronze:** about 90% copper and 10% tin; hard and corrosion-resistant, used for statues, coins, bells and bearings.

Organic Compounds

UNIT 3

Sources

LESSON 12

1. Organic Chemistry and the Vital Force Theory

Organic chemistry is the branch of chemistry that deals with carbon compounds, mainly hydrocarbons and their derivatives. The vital force theory stated that organic compounds could only be produced inside living organisms with the help of a mysterious vital force, and could never be prepared in the laboratory. It was disproved in 1828 when Friedrich Wohler prepared urea, a typical organic compound, by heating the inorganic salt ammonium cyanate: NH4CNO -> CO(NH2)2. After this, many organic compounds were synthesised in the laboratory, so the theory was abandoned.

2. Natural Sources of Organic Compounds

The main natural sources are fossil remains, plants and animals. Coal, petroleum and natural gas were formed from the remains of plants and animals buried millions of years ago under high temperature and pressure, and they supply most hydrocarbons and fuels. Natural gas consists mainly of methane, usually about 85-90%, with small amounts of ethane, propane and butane, so methane is its chief constituent and the main source of its fuel value.

3. Plants and Animals as Sources

Natural product chemistry deals with the isolation, structure and synthesis of organic compounds produced by living organisms. Plants are a rich source: they give carbohydrates, vegetable oils, essential oils, alkaloids such as quinine and morphine, natural rubber, and vitamins. Animals provide fats and oils, proteins, hormones such as insulin, enzymes, wool and silk. Many important medicines were first obtained from these natural sources and are now also made in the laboratory.

4. Total and Partial Synthesis

In total synthesis an organic compound is built up in the laboratory starting from the elements themselves or from very simple inorganic materials, passing through all the steps in the laboratory. In partial synthesis the starting material is a natural organic compound obtained from plants or animals, which is then converted into the desired product by a few chemical steps. Total synthesis proves the structure of a compound completely, while partial synthesis is shorter and is often used commercially.

Coal as a Source of Organic Compounds

LESSON 13

1. Destructive Distillation of Coal

Destructive distillation is the heating of coal strongly in the absence of air, so that it decomposes instead of burning. The products are: coke, the solid residue used as a fuel and reducing agent in metallurgy; coal tar, a thick black liquid from which benzene, toluene, phenol, naphthalene and many dyes and drugs are obtained; coal gas, a mixture of hydrogen, methane and carbon monoxide used as a fuel; and ammonia liquor (ammoniacal liquor), an aqueous solution containing ammonia formed from the nitrogen present in coal, used to make fertilisers such as ammonium sulphate. The process is carried out industrially in coke ovens and gas works.

2. Types of Coal

Coal is classified by rank, that is, by increasing carbon content and calorific value, into peat, lignite, bituminous coal and anthracite.

- **Peat:** the first stage, soft and brown, with the least carbon (about 30-40% in dry form) and much moisture.
- **Lignite (brown coal):** contains more carbon and less moisture than peat.
- **Bituminous coal:** black, contains about 70-85% carbon and is the most widely used coal for industry and power generation.

- **Anthracite:** the highest-rank coal, hard and shiny, with about 90% or more carbon, the highest calorific value and a clean, almost smokeless flame.

Characteristics of Organic Compounds

LESSON 14

1. Unique Characteristics

Organic compounds have several unique features. Carbon shows catenation and forms chains, branches and rings, and it forms strong covalent bonds with H, O, N, S and halogens, giving millions of compounds. They show isomerism, are generally covalent with low melting and boiling points, and are mostly insoluble in water but soluble in organic solvents. Their reactions are usually slow and molecular rather than ionic, and most of them are combustible.

2. Catenation

Catenation is the property of an element, especially carbon, to form long chains and rings by joining with atoms of the same element through covalent bonds. Carbon shows catenation to the maximum extent because the C-C bond is strong and carbon is small and tetravalent. This property is a main reason for the existence of millions of organic compounds.

3. Isomerism

Isomerism is the phenomenon in which two or more compounds have the same molecular formula but different structures and therefore different properties; such compounds are called isomers. For example, n-butane and isobutane both have the formula C4H10 but different arrangements of carbon atoms: CH3-CH2-CH2-CH3 and CH3-CH(CH3)-CH3 are a pair of structural isomers. Isomerism may be structural, based on different connectivity, or stereo, based on different spatial arrangement. Pairs with different molecular formulas, such as C2H5-OH and C3H7-OH, are homologues, not isomers.

Uses of Organic Compounds

LESSON 15

1. Importance in Everyday Life

Organic compounds are essential in daily life: fuels such as petrol, LPG and natural gas, medicines such as aspirin and antibiotics, plastics, synthetic fibres like nylon, soaps, detergents, dyes, perfumes and food items are all organic. Common uses include fuels (petrol, diesel, natural gas, coal), medicines (aspirin, paracetamol, antibiotics), polymers (polythene, PVC, nylon, polyester) for plastics and fibres, and solvents, paints, dyes, perfumes, soaps, detergents, pesticides and fertilisers such as urea. Almost every industry from agriculture to medicine depends on organic compounds.

2. Reforming of Petrol

Reforming is the process in which straight-chain alkanes of low-grade petrol are converted into branched-chain alkanes and aromatic hydrocarbons by heating with a catalyst such as platinum. It is useful because branched and aromatic hydrocarbons have a high octane number and burn smoothly without knocking, the damaging metallic sound produced by uneven combustion in the engine. Reforming therefore improves the quality of petrol for modern high-compression engines.

3. Products of Biotechnology

Biotechnology produces useful materials using living organisms or enzymes. Examples include insulin for diabetic patients, antibiotics such as penicillin, vaccines, interferons and growth hormones. It is also used to produce ethanol fuel, enzymes for detergents, and improved seeds and crops in agriculture.

New Allotrope of Carbon: Bucky Ball

LESSON 16

1. Buckminsterfullerene, C60

Bucky ball is the common name of buckminsterfullerene, C60, an allotrope of carbon discovered in 1985. Its 60 carbon atoms form a hollow cage shaped like a football, made of exactly 12 pentagonal (five-membered) rings and 20 hexagonal (six-membered) rings, with each carbon sp2 hybridised and part of a delocalised pi system.

2. Properties and Uses

It is a black solid, insoluble in water but soluble in organic solvents such as benzene, and it can accept electrons to form salts. Fullerenes are used in research on superconductors, lubricants, drug delivery and new materials.

Functional Groups and Homologous Series

LESSON 17

1. Functional Groups

A functional group is an atom or a group of atoms in a molecule which is responsible for the characteristic chemical properties of that compound. The amide group -CONH2 is the functional group that contains a nitrogen atom bonded to a carbonyl carbon; alcohol (-OH), ketone (C=O) and ester (-COO-) groups contain only carbon, hydrogen and oxygen. Three oxygen-containing functional groups are: hydroxyl group -OH (alcohol, R-OH); carbonyl group C=O, in aldehydes as -CHO (R-CHO) and in ketones as R-CO-R; and carboxyl group -COOH (carboxylic acid, R-COOH). The ester group -COO- (R-COO-R) is another common oxygen-containing group.

2. Homologous Series

A homologous series is a family of organic compounds having the same functional group and similar chemical properties, in which successive members differ by a -CH2- group, for example the alkanes CH4, C2H6, C3H8 and so on. Its main properties are: all members can be represented by one general formula, such as CnH2n+2 for alkanes and CnH2n+1OH (or R-OH) for alcohols, examples being CH3OH (methanol) and C2H5OH (ethanol); all members show similar chemical properties because they have the same functional group; physical properties such as boiling point change gradually with increasing molecular mass; and all members can be prepared by similar general methods. C2H4 (ethene) belongs to the alkene family because it fits the alkene general formula CnH2n and has a carbon-carbon double bond, while CH4, C2H6 and C3H8 are alkanes of formula CnH2n+2.

Nomenclature of Organic Compounds

UNIT 4

Introduction of Hydrocarbons and their Derivatives

LESSON 18

Organic compounds exhibit great diversity, with millions of known compounds and an infinite number of possible isomers. They can possess distinct functional groups, different types of substituents, and variable lengths, branches and cyclic structures of carbon chains, leading to a wide range of structural possibilities. It was a big challenge for organic chemists to establish a systematic and simplified method for naming this vast number of compounds in a consistent and uniform manner.

1. Hydrocarbons and their Derivatives

Organic compounds are mainly made up of carbon and hydrogen; certain other elements such as oxygen, nitrogen, sulphur and halogens may also be present in their structures. A wide range of organic compounds are derivatives of hydrocarbons, formed by substituting one or more hydrogen atoms with a hetero atom or group of atoms. These derivatives are distinguished by their unique structures and properties and are categorised into families or groups of organic compounds. Each family is characterised by a specific functional group, which imparts distinct chemical and physical properties to the compounds within that family. Functional groups are specific groups of atoms that are responsible for the characteristic chemical reactions and properties of organic compounds.

2. Five Organic Families

- **Alcohols:** functional group -OH (hydroxyl); suffix -ol; for example ethanol.
- **Aldehydes:** functional group -CHO; suffix -al; for example ethanal.
- **Ketones:** functional group C=O (carbonyl) within the chain; suffix -one; for example propanone.
- **Carboxylic acids:** functional group -COOH; suffix -oic acid; for example ethanoic acid.
- **Amines:** functional group -NH2; suffix -amine; for example methylamine.

3. Root Words for the Carbon Chain

The prefix (root word) shows the number of carbon atoms in the chain: 1 Meth-, 2 Eth-, 3 Prop-, 4 But-, 5 Pent-, 6 Hex-, 7 Hept-, 8 Oct-, 9 Non-, 10 Dec-, 11 Undec-, 12 Dodec-. With the primary suffixes these give methane, ethane, propane and so on for alkanes; ethene, propene, butene for alkenes; and ethyne, propyne, butyne for alkynes.

History of Nomenclature

LESSON 19

1. Common (Trivial) Names

Initially the names of organic compounds were assigned on the basis of their origin or the person who discovered them. These specific designations are referred to as common names or trivial names. With the rapid growth of organic chemistry the number of compounds increased tremendously, and it became impossible to assign a common name to such an extensive range of compounds; scientists felt the need for a structural or systematic name that would accurately reflect the molecular structure of the compound. The common system gives names based on source or history, so one compound may have several names and the name tells nothing about its structure.

2. Geneva Conference and IUPAC

An attempt was made at the Geneva conference (1892) to create a unique nomenclature in which only one official name would be assigned to each compound. In 1958 a new commission on nomenclature was appointed by the International Union of Pure and Applied Chemistry (IUPAC) for further work on the nomenclature, and the system has set rules for naming organic molecules on the basis of their structures. The latest updates to the IUPAC rules were recommended in 2013.

3. Iso and Neo Prefixes of the Common System

In the common system, the prefix iso- is used when one methyl group is attached to the second carbon of the chain, that is, the chain ends in a -CH(CH3)2 group; for example, (CH3)2CH-CH3 is isobutane and (CH3)2CH-CH2-OH is isobutyl alcohol. The prefix neo- is used when two methyl groups are attached to the second carbon, that is, the chain contains a -C(CH3)3 end group; for example, (CH3)4C is neopentane. These prefixes describe branching without using numbers.

IUPAC System

LESSON 20

The systematic or scientific system for the naming of organic compounds is the IUPAC method. This system provides distinct names for over fifteen million reported organic compounds and can be devised for the millions of compounds yet to be synthesised. The IUPAC method considers the arrangement of carbon atoms, the types and positions of functional groups and other distinctive features of the compound to generate a standard name. An IUPAC name is built from a prefix (substituents), a root word (length of the carbon chain), a primary suffix (nature of the carbon chain: -ane, -ene, -yne) and a secondary suffix (functional group).

1. General Rules

- Identify the longest unbroken (continuous) carbon chain within the molecule which will act as the parent chain, selecting the chain that contains the functional group.
- Allocate numerical designations (locants) to the carbon atoms of the parent chain, numbering from the terminal side to which the functional group is nearer, so that the functional group and then the substituents get the lowest possible locants.
- Identify substituents and name them using prefixes such as methyl, ethyl, methoxy, chloro, bromo, etc., mentioning their position in the longest carbon chain. Substituent prefixes are written in alphabetical order, using di-, tri- for repeats.
- If any functional group is present in the molecule, name it by using the appropriate suffix, such as -one for ketone, -al for aldehyde and -ol for alcohol, etc.
- If more than one functional group is present, write the name on the priority basis set by the priority rules: (-COOH) > (-CHO) > (C=O) > (-OH) > (-NH2) > (-O-) > (=) > (triple bond). The highest-priority group becomes the suffix and the others are named as prefixes.

Locants are separated by commas and are joined to the name by hyphens. For example, CH3-CH(CH3)-CH2-OH is 2-methylpropan-1-ol. The IUPAC system is preferred over the common system because it is systematic and universal: each compound gets one unique name from its structure, and from the name the correct structure can be drawn anywhere in the world.

Nomenclature of Alkane

LESSON 21

1. Common System

In the common system of naming alkanes, all the carbon atoms present in the structure are included in the name, regardless of whether they are attached to each other in a straight chain or in a branched form; the prefixes iso- and neo- are used for branched chains (see History of Nomenclature). In the IUPAC system the longest continuous chain is selected as the parent and branches are named as alkyl prefixes with locants.

2. Cycloalkanes

An open-chain saturated hydrocarbon must fit the formula CnH2n+2, as butane is C4H10. A saturated hydrocarbon with the formula C4H8 can only be saturated if it is cyclic, because forming the ring removes two hydrogen atoms; C4H8 as a saturated compound is therefore cyclobutane. Ring compounds are named by adding the prefix cyclo- to the name of the corresponding open-chain alkane.

Nomenclature of Alkene

LESSON 22

Alkenes are named by replacing the suffix -ane of the parent alkane with -ene, and the position of the double bond is shown by the locant of its first carbon. The structure of 1,3-pentadiene is CH2=CH-CH=CH-CH3: the two double bonds start at carbons 1 and 3 of the five-carbon chain. (CH2=CH-CH2-CH=CH2 would be 1,4-pentadiene.)

1. Chains with Both Double and Triple Bonds

If an organic compound contains both a double and a triple bond in the main carbon chain, the chain is numbered so that the multiple bonds together get the lowest possible locants; if there is still a choice, the double bond is given the lower locant. In the suffix, -ene is written before -yne, and the final -e of -ene is dropped, giving endings like -en-yne with both locants stated. For example, HC triple bond C-CH=CH2 is but-3-en-1-yne, and CH3-CH=CH-C triple bond CH is pent-3-en-1-yne: in the latter, the chain of five carbons is numbered from the end nearer the triple bond, giving the triple bond position 1 and the double bond position 3.

2. Worked Example

HC triple bond C-CH(CH3)-CH=CH2: the longest chain has five carbons and contains both a triple bond and a double bond, with a methyl branch on carbon 3. Numbering from the triple-bond end gives the triple bond position 1 and the double bond position 4. Its IUPAC name is 3-methylpent-4-en-1-yne.

Nomenclature of Alkyne

LESSON 23

Alkynes are named by replacing the suffix -ane of the parent alkane with -yne, with a locant for the triple bond where needed. A hydrocarbon with the molecular formula C7H12 fits the alkyne general formula CnH2n-2 (for n = 7, 2n - 2 = 12), so it is heptyne with one triple bond; heptane is C7H16 and heptene is C7H14, so they do not fit this formula. When a chain contains both a double and a triple bond, the name ends in -en-yne and the numbering rules of the alkene lesson apply, for example pent-3-en-1-yne and 3-methylpent-4-en-1-yne.

Nomenclature of Alkyl Halide

LESSON 24

Alkyl halides are the derivatives of alkanes in which one or more hydrogen atoms are replaced by a halogen atom.

1. Common System

The common name of an alkyl halide is formed by naming the alkyl group first, followed by the term "halide": CH3-Cl is methyl chloride, CH3-CH2-Br is ethyl bromide and CH3-CH2-CH2-I is propyl iodide. The terms secondary (sec.) and tertiary (tert.) are used as prefixes if the carbon atom bonded to halogen is further attached to two and three other carbon atoms respectively, for example sec-propyl chloride, CH3-CH(Cl)-CH3, and tert-butyl bromide, (CH3)3C-Br. The terms iso and neo are also used as prefixes when the penultimate carbon is further attached to two or three other carbon atoms respectively, for example isobutyl bromide and neopentyl iodide.

2. IUPAC System

In the IUPAC system alkyl halides are named as haloalkanes, where halo is chloro, bromo, iodo, etc. The rules are: (i) determine the longest carbon chain that contains the carbon atom directly bonded to halogen; (ii) number the carbon atoms in the chain from the end closest to the halogen atom; (iii) when a double or triple bond is present in the halogenated chain, number from the end nearest to the double or triple bond; (iv) prefixes di-, tri-, tetra, etc. indicate the number of halogen atoms attached to the main chain; (v) when two different halogens are attached at the same position from opposite ends, the halogen names are written in alphabetical order. Examples: CH3-CH2-Br is bromoethane; CH3-CH(Cl)-CH3 is 2-chloropropane; (CH3)3C-Br is 2-bromo-2-methylpropane; and a five-carbon chain with Br at position 2 and Cl at position 4 is 2-bromo-4-chloropentane. Dichlorodifluoromethane (CCl2F2) has been used as a refrigerant gas; it is now being replaced by more environmentally friendly alternatives.

Nomenclature of Amines

LESSON 25

Amines are organic compounds that contain a nitrogen atom bonded to one or more alkyl or aryl groups. A primary amine contains one alkyl group bonded to the nitrogen atom, while secondary and tertiary amines are identified by the direct attachment of two and three alkyl groups to the nitrogen atom respectively. In the IUPAC system the suffix for the amine family is -amine, added to the name of the parent chain, for example methylamine and ethylamine from the -NH2 functional group.

1. Worked Example: 3-Chlorobenzamide

Benzamide is a benzene ring attached to a -CONH2 group, which is taken as position 1; in 3-chlorobenzamide chlorine is placed at position 3 (meta). Its structure is written as Cl-C6H4-CONH2 with Cl meta to the amide group. Amides are the nitrogen derivatives of carboxylic acids.

Nomenclature of Alcohol

LESSON 26

Alcohols are named by replacing the final -e of the parent alkane with the suffix -ol, with a locant for the -OH group where needed. The IUPAC name of isopropyl alcohol is 2-propanol (propan-2-ol): isopropyl alcohol is (CH3)2CH-OH, in which the -OH group is attached to the middle carbon of a three-carbon chain, so numbering gives the -OH group position 2.

1. Worked Example: Glycerol

CH2(OH)-CH(OH)-CH2(OH) has a three-carbon chain with one hydroxyl group on every carbon. Its IUPAC name is propane-1,2,3-triol. Its common name is glycerol (glycerine), obtained from fats and used in medicines, cosmetics and food.

Nomenclature of Phenol

LESSON 27

Phenol is the parent name for a benzene ring carrying an -OH group, and the ring is numbered from the carbon bearing the -OH group. Resorcinol is a phenol containing two -OH groups at positions 1 and 3; it is benzene-1,3-diol, with the two -OH groups in meta positions to each other. The 1,2-diol is catechol and the 1,4-diol is hydroquinone (quinol). 1,2,3-Benzenetriol is a ring carrying three -OH groups on three consecutive carbons; its common name is pyrogallol, used in photography and hair dyes.

1. Worked Examples

4-Ethylphenol: a benzene ring with -OH at the top and -C2H5 at the bottom (para positions). The -OH group makes the parent compound phenol, and the ethyl group is opposite to it, at position 4, so the name is 4-ethylphenol.

Picric acid: picric acid is 2,4,6-trinitrophenol: a phenol ring carrying three -NO2 groups at positions 2, 4 and 6 around the -OH group, C6H2(NO2)3-OH. The three electron-withdrawing nitro groups make it a strong acid for a phenol, and it is also an explosive and a yellow dye.

Nomenclature of Ether

LESSON 28

Ethers contain an oxygen atom joining two alkyl or aryl groups, R-O-R. They are named either as alkoxyalkanes in the substitutive system or by naming the two groups followed by the word ether in the common functional-class system.

1. Worked Examples

CH3-CH2-O-C(CH3)2-C2H5: oxygen joins an ethyl group and a C(CH3)2(C2H5) group. Treating it as a substituted butane, the longest chain is four carbons with an ethoxy group and a methyl group on carbon 2. Its IUPAC name is 2-ethoxy-2-methylbutane, also named ethyl (2-methylbutan-2-yl) ether in the functional-class system.

Ethyl neopentyl ether: neopentyl is the group (CH3)3C-CH2- and ethyl is CH3CH2-, joined through an oxygen atom: CH3CH2-O-CH2-C(CH3)3. It is an unsymmetrical ether because the two alkyl groups on oxygen are different.

Nomenclature of Aldehyde and Ketone

LESSON 29

Aldehydes are named with the suffix -al, the aldehyde carbon always being position 1. For ketones the final -e of the main carbon chain is replaced by the suffix -one, as in propanone and butan-2-one. (The suffix -ol is used for alcohols and -al for aldehydes.)

1. Worked Examples: Naming Structures

- **(CH3)2CH-C(=O)-CH(Cl)-CH3:** the longest chain containing the carbonyl group has five carbons, with C=O at carbon 3, a methyl group at carbon 2 and a chlorine at carbon 4: 4-chloro-2-methylpentan-3-one. Substituent prefixes are written in alphabetical order (chloro before methyl) with their locants.
- **(CH3)2CBr-CHO:** the -CHO group is carbon 1 of a three-carbon chain, and carbon 2 carries both a bromine atom and a methyl group: 2-bromo-2-methylpropanal.
- **CH3-C(=O)-CH=CH-CH3:** five carbons with the ketone group at carbon 2 and a double bond between carbons 3 and 4; numbering gives the carbonyl group the lower locant: pent-3-en-2-one.

- **Benzene ring with -COCH3 and -Cl in meta positions:** the -COCH3 (acetyl) group is taken as position 1 and the chlorine is at position 3 (meta). The name is 1-(3-chlorophenyl)ethan-1-one, commonly called 3-chloroacetophenone; acetophenone itself is named systematically as 1-phenylethan-1-one.
- **1-Bromo-3-methylhexan-2-one:** a six-carbon chain with a ketone group at carbon 2, bromine at carbon 1 and a methyl branch at carbon 3: BrCH2-C(=O)-CH(CH3)-CH2-CH2-CH3.

2. Worked Examples: Structures from Names

- **Butyraldehyde:** the common name of the four-carbon aldehyde with a straight chain ending in -CHO: CH3-CH2-CH2-CHO. Its IUPAC name is butanal.
- **Ethyl isopropyl ketone:** in the common ketone naming system the two groups on the carbonyl carbon are ethyl (CH3CH2-) and isopropyl ((CH3)2CH-): CH3-CH2-C(=O)-CH(CH3)2. Its IUPAC name is 2-methylpentan-3-one.
- **alpha-Methyl butyraldehyde:** butyraldehyde is CH3-CH2-CH2-CHO; the alpha carbon is the one next to the -CHO group and it carries a methyl group: CH3-CH2-CH(CH3)-CHO, IUPAC name 2-methylbutanal.

Nomenclature of Carboxylic Acid

LESSON 30

Carboxylic acids are named by replacing the final -e of the parent alkane with -oic acid, the -COOH carbon being carbon 1. In the common system the straight-chain six-carbon acid CH3-(CH2)4-COOH is called caproic acid (IUPAC name hexanoic acid).

1. Common Names of the First Five Acids

- **HCOOH:** formic acid (1 carbon) - IUPAC methanoic acid
- **CH3COOH:** acetic acid (2 carbons) - IUPAC ethanoic acid
- **CH3CH2COOH:** propionic acid (3 carbons) - IUPAC propanoic acid
- **CH3CH2CH2COOH:** butyric acid (4 carbons) - IUPAC butanoic acid
- **CH3(CH2)3COOH:** valeric acid (5 carbons) - IUPAC pentanoic acid

The common names come from natural sources: formic acid from ants, acetic acid from vinegar, and caproic acid is the six-carbon acid (propionic has three carbons, valeric five and stearic eighteen).

2. Worked Examples

- **CH2=C(CH3)-COOH:** the parent chain has three carbons including the -COOH carbon, which is carbon 1, with a double bond between carbons 2 and 3 and a methyl group on carbon 2: 2-methylprop-2-enoic acid. Its common name is methacrylic acid, an important monomer for plastics.
- **alpha,beta-Dimethyl butyric acid:** butyric acid is a four-carbon acid; the carbon next to -COOH is alpha and the next is beta, and each carries one methyl group: CH3-CH(CH3)-CH(CH3)-COOH, in IUPAC terms 2,3-dimethylbutanoic acid.
- **alpha-Chloro-beta-methyl valeric acid:** valeric acid is a five-carbon acid; chlorine is on the alpha carbon (position 2) and a methyl group on the beta carbon (position 3): CH3-CH2-CH(CH3)-CH(Cl)-COOH, in IUPAC terms 2-chloro-3-methylpentanoic acid.

Nomenclature of Ester

LESSON 31

An ester is named as alkyl alkanoate: the alkyl part from the alcohol is written first, followed by the name of the acid part with -oate ending. CH3COOCH(CH3)2 is therefore isopropyl ethanoate: the acid part CH3COO- comes from ethanoic acid and the alcohol part -CH(CH3)2 is the isopropyl group. Isopropyl acetate is only its common name, since acetate is the common name of ethanoate.

1. Worked Examples

- **CH3-C(=O)-O-CH(CH3)2:** an ester formed from ethanoic acid and propan-2-ol, because the -O-CH(CH3)2 part is the isopropyl (propan-2-yl) group: isopropyl ethanoate (propan-2-yl ethanoate).
- **Isopropyl butanoate:** formed from butanoic acid and propan-2-ol, so the acid part is CH3-CH2-CH2-COO- and the alcohol part is -CH(CH3)2: CH3-CH2-CH2-C(=O)-O-CH(CH3)2. Like other low-molecular esters it has a fruity smell.

Nomenclature of Amide

LESSON 32

Amides are the nitrogen derivatives of carboxylic acids, in which the -OH of the -COOH group is replaced by -NH2; they are named from the parent acid by changing -oic acid to -amide. The worked example of this family is 3-chlorobenzamide: benzamide is a benzene ring attached to a -CONH2 group, taken as position 1, with chlorine at position 3 (meta) - Cl-C6H4-CONH2 with Cl meta to the amide group.

Nomenclature of Acyl Halides

LESSON 33

The acyl group is R-CO-, and in R-CO-X it is attached to a halogen atom, forming an acyl halide, the typical acyl compound. (R-CO-R is a ketone, R-COO-CH3 is an ester, and in an amide the acyl group is present as part of the -CONH2 group.) Acyl halides are named from the parent acid by changing -oic acid to -oyl halide.

1. Worked Example: Butanoyl Iodide

Butanoyl iodide is the acyl halide of butanoic acid, with the -OH of the acid replaced by iodine: CH3-CH2-CH2-C(=O)-I. Acyl halides are highly reactive and fume in moist air.

Hydrocarbons

UNIT 5

Types of Hydrocarbons

LESSON 34

1. Aliphatic and Aromatic Hydrocarbons

- Aliphatic hydrocarbons have open-chain or non-aromatic ring structures, while aromatic hydrocarbons contain a benzene ring with delocalised pi electrons.
- Aliphatic unsaturated compounds mainly give addition reactions, while aromatic compounds prefer electrophilic substitution reactions that preserve the ring.
- Aromatic hydrocarbons have a much higher carbon percentage and burn with a smoky flame, whereas most aliphatic hydrocarbons burn with a comparatively clean flame.

2. Reactivity: Ethane, Ethene and Ethyne Compared

Reactivity increases with the number of pi bonds, because pi electrons are loosely held, exposed and easily attacked by electrophiles, whereas the sigma bonds of ethane are strong and non-polar. Ethane has no pi bond and reacts only by slow free-radical substitution. Ethene has one pi bond and readily gives addition reactions, while ethyne has two pi bonds and an even greater density of exposed pi electrons, so it is the most reactive of the three towards addition: ethyne > ethene > ethane.

Alkanes and Cycloalkanes

LESSON 35

1. Why Alkanes are Called Paraffins

The word paraffin means little affinity, and alkanes are unreactive because all their bonds are strong, non-polar C-C and C-H sigma bonds. They have no pi bonds, lone pairs or polar sites for reagents to attack, and their bond energies are high. They therefore do not react with acids, alkalis, oxidising agents or electrophiles under ordinary conditions, and react mainly by free-radical substitution in sunlight.

2. Orbital Structure of Ethane

In ethane each carbon is sp3 hybridised and tetrahedral, with bond angles of about 109.5 degrees. The C-C bond is formed by head-on overlap of two sp3 orbitals (a sigma bond), and each C-H bond is formed by overlap of an sp3 orbital of carbon with the 1s orbital of hydrogen. The structure CH3-CH3 therefore contains seven sigma bonds, and free rotation is possible about the C-C bond.

3. Cycloalkanes

A saturated hydrocarbon with the formula C3H6 cannot be propane (C3H8) or propene (C3H6 is unsaturated) or propyne (C3H4); a saturated compound of this formula is only possible as a ring, cyclopropane, because ring formation removes two hydrogen atoms. Cycloalkanes fit the general formula CnH2n.

Radical Substitution Reactions

LESSON 36

1. Free Radical Chlorination of Methane

The reaction of methane with chlorine in sunlight occurs in three stages.

- **Initiation:** sunlight splits a chlorine molecule into two free radicals: Cl2 -> 2Cl. (in the presence of hv).
- **Propagation:** Cl. + CH4 -> CH3. + HCl, and then CH3. + Cl2 -> CH3Cl + Cl., so the chain continues and more chlorine can be substituted to give CH2Cl2, CHCl3 and CCl4.
- **Termination:** two radicals combine and end the chain, for example Cl. + Cl. -> Cl2 and CH3. + CH3. -> C2H6.

Alkenes

LESSON 37

1. Orbital Structure of Ethylene

In ethylene (ethene) each carbon is sp2 hybridised and the molecule is planar with bond angles of about 120 degrees. The C-C sigma bond is formed by overlap of sp2 orbitals, the four C-H bonds by sp2-s overlap, and the second C-C bond is a pi bond formed by sideways overlap of the unhybridised p orbitals above and below the plane. Ethene thus has one sigma and one pi bond between the carbons, CH2=CH2, and rotation about the double bond is restricted.

2. Important Reactions of Alkenes

- **Dehydration of ethanol:** C2H5OH -> CH2=CH2 + H2O (conc. H2SO4, 170 C). Ethanol is dehydrated by concentrated sulphuric acid, -OH being removed from one carbon and H from the adjacent carbon (elimination); the major product is ethene.
- **Dehydrohalogenation:** C2H5Br + KOH (alcoholic, heat) -> CH2=CH2 + KBr + H2O. With alcoholic KOH ethyl bromide loses HBr to give ethene; aqueous KOH would instead give ethanol by substitution.
- **Addition of HBr (Markownikov rule):** CH3-CH=CH2 + HBr -> CH3-CHBr-CH3. The hydrogen goes to the carbon already having more hydrogens and bromine attaches to the middle carbon; the major product is 2-bromopropane.
- **Ozonolysis:** CH2=CH2 + O3 -> (ethene ozonide) -> 2HCHO (with Zn/H2O). The ozonide on decomposition with Zn and water cleaves the double bond and gives carbonyl compounds; aldehydes are formed from =CH- groups, so ethene gives formaldehyde (methanal).

3. Polymerisation and Epoxidation

On heating under high pressure with a peroxide catalyst, ethene molecules join together to form the polymer polythene: n(CH2=CH2) -> -(CH2-CH2)n-. This is addition polymerisation, and polythene is used for bags, bottles and insulation. In the presence of peracetic acid, ethene is oxidised (epoxidised) to ethylene oxide, a three-membered cyclic ether: CH2=CH2 + CH3COOOH -> C2H4O (ethylene oxide) + CH3COOH. Ethylene oxide is an important industrial chemical used to make ethylene glycol and detergents.

Alkynes

LESSON 38

1. Orbital Structure of Acetylene

In acetylene (ethyne) each carbon is sp hybridised and the molecule is linear with a bond angle of 180 degrees. The C-C sigma bond comes from sp-sp overlap, the two C-H bonds from sp-s overlap, and two pi bonds are formed by sideways overlap of the two pairs of unhybridised p orbitals. Ethyne, HC triple bond CH, therefore contains one sigma and two pi bonds between the carbon atoms. Acetylene burns in oxygen with an extremely hot flame of about 3000 C, called the oxy-acetylene flame, which is hot enough to melt and join metals, so acetylene is the welding gas.

2. Addition Reactions

- **Addition of HBr:** ethyne adds two molecules of HBr in two steps, and both additions follow Markownikov rule, so both bromine atoms go to the same carbon. The final product is CH3-CHBr2, 1,1-dibromoethane; bromoethene is only the intermediate of the first step.
- **Addition of chlorine:** HC triple bond CH + Cl2 -> CHCl=CHCl (1,2-dichloroethene), and CHCl=CHCl + Cl2 -> CHCl2-CHCl2. The final product, 1,1,2,2-tetrachloroethane, is a heavy liquid formerly used as a solvent.

3. Preparations

Ethyne from ethyl chloride: ethyl chloride is treated with alcoholic KOH to give ethene (CH3CH2Cl -> CH2=CH2), brominated to 1,2-dibromoethane (CH2Br-CH2Br), and then heated with strong alcoholic KOH (or NaNH2), which removes 2HBr: CH2Br-CH2Br -> HC triple bond CH + 2HBr. In the same way, 1,2-dibromobutane with 2KOH (alc., heat) gives but-1-yne: CH3-CH2-CHBr-CH2Br -> CH3-CH2-C triple bond CH + 2KBr + 2H2O. Heating 1,1,2,2-tetrabromoethane with zinc powder removes bromine (debromination) and forms the triple bond: CHBr2-CHBr2 + 2Zn -> HC triple bond CH + 2ZnBr2.

4. Acidic Nature of Terminal Alkynes

In ethyne the carbon is sp hybridised and has 50% s-character, so it is strongly electronegative and holds the C-H electrons close, allowing the hydrogen to leave as H+ and leaving a stable acetylide ion. Two reactions show this acidity: (1) with sodium metal: 2HC triple bond CH + 2Na -> 2HC triple bond CNa (sodium acetylide) + H2; (2) with ammoniacal silver nitrate it gives a white precipitate of silver acetylide: HC triple bond CH + 2[Ag(NH3)2]OH -> AgC triple bond CAg + 4NH3 + 2H2O. This test distinguishes ethyne from ethene, which gives no such precipitate, although both gases decolourise bromine water and KMnO4 solution. Internal alkynes and alkenes do not show these reactions.

Isomerism

LESSON 39

Isomerism is the existence of compounds with the same molecular formula but different structures and properties.

1. Types of Structural Isomerism

- **Chain isomerism:** due to different carbon skeletons, for example n-butane and isobutane (C4H10).
- **Position isomerism:** due to different positions of a functional group or multiple bond, for example 1-butene and 2-butene.
- **Functional group isomerism:** due to different functional groups, for example ethanol and dimethyl ether (both C2H6O; ethanol has the -OH alcohol group and dimethyl ether the -O- ether group).
- **Metamerism:** due to different alkyl groups on either side of a polyvalent functional group, for example diethyl ether and methyl propyl ether (C4H10O).

2. Chiral Carbon and Optical Isomerism

A chiral carbon is a carbon atom attached to four different atoms or groups, so that its molecule has no plane of symmetry and exists as non-superimposable mirror images (optical isomers). An example is the central carbon of lactic acid, CH3-CH(OH)-COOH, which is bonded to -H, -OH, -CH3 and -COOH, four different groups. Optical isomers (enantiomers) have identical physical and chemical properties except that they rotate plane-polarised light in opposite directions; lactic acid exists as (+) and (-) forms, and 2-butanol is another example pair. Such a compound shows optical activity.

3. Geometrical Isomerism

Geometrical (cis-trans) isomers are stereoisomers arising from restricted rotation about a double bond, in which the groups attached to the double-bonded carbons are arranged differently in space. In cis-2-butene both -CH3 groups are on the same side of the double bond, while in trans-2-butene they are on opposite sides. The two isomers have different melting points, boiling points and dipole moments.

Benzene and its Derivatives

LESSON 40

1. Molecular Orbital Structure of Benzene

In benzene all six carbons are sp2 hybridised and form a planar hexagonal ring with bond angles of 120 degrees; the C-C sigma framework is made from sp2-sp2 overlaps and each C-H bond from sp2-s overlap. Each carbon has one unhybridised p orbital perpendicular to the ring, and the six p orbitals overlap sideways all around the ring to form a continuous delocalised pi cloud above and below the ring plane. This delocalisation makes all C-C bonds equal in length (between single and double bond values) and gives benzene its extra stability (resonance energy about 150 kJ/mol) and substitution rather than addition reactions. Benzene, C6H6, contains about 92% carbon, and there is not enough air in an ordinary flame to burn all of it completely; the unburnt carbon particles glow and escape as soot, so benzene burns with a smoky flame.

2. Reactions of Benzene

- **Hydrogenation:** C6H6 + 3H2 -> C6H12 (Ni catalyst, about 150-200 C, pressure). All three pi bonds of the ring are reduced to give cyclohexane.
- **Acylation (Friedel-Crafts):** C6H6 + CH3COCl -> C6H5COCH3 + HCl (anhydrous AlCl3). The product is acetophenone, an aromatic ketone. Reaction with CH3Cl under similar conditions (alkylation) would give toluene.
- **Nitration:** C6H6 + HNO3 -> C6H5NO2 + H2O (conc. H2SO4, 50-60 C). Nitrobenzene is a pale yellow oily liquid smelling of bitter almonds.
- **Sulphonation:** C6H6 + H2SO4 -> C6H5SO3H + H2O (conc./fuming H2SO4, heat). The reaction is reversible and is an electrophilic substitution by the SO3 electrophile.

3. Mechanism of Electrophilic Substitution

Nitration: (1) conc. HNO3 + conc. H2SO4 generate the nitronium ion, NO2+, the attacking electrophile (HNO3 + 2H2SO4 -> NO2+ + H3O+ + 2HSO4-). (2) The pi electrons of benzene attack NO2+ to form a resonance-stabilised carbocation (arenium ion) intermediate, in which aromaticity is temporarily lost. (3) HSO4- removes the proton from the carbon bearing the NO2 group, the pi system is restored and nitrobenzene is formed.

Acylation: (1) acetyl chloride reacts with anhydrous AlCl3 to form the acylium ion electrophile, CH3CO+ (CH3COCl + AlCl3 -> CH3CO+ + AlCl4-). (2) The benzene pi cloud attacks the acylium ion to give a resonance-stabilised arenium ion. (3) AlCl4- removes the ring proton, aromaticity is restored and acetophenone is formed, with AlCl3 regenerated on aqueous work-up.

Chlorination: (1) chlorine reacts with the Lewis acid catalyst FeCl3 (or AlCl3) to generate the electrophile Cl+ (Cl2 + FeCl3 -> Cl+ + FeCl4-). (2) Benzene pi electrons attack Cl+ to form a resonance-stabilised arenium ion. (3) FeCl4- removes the proton from the same carbon, the aromatic pi system is restored, and chlorobenzene plus HCl are formed with the catalyst regenerated.

4. Directing Groups

Positions 2 and 6 relative to a substituent are ortho, 3 and 5 are meta, and 4 is para. Activating groups with lone pairs or electron-donating effect, such as -OH, -NH2 and -CH3, increase electron density at the ortho and para positions, so they direct the incoming electrophile there and speed up substitution (o/p directors). Deactivating electron-withdrawing groups, such as -NO2, -COOH and -CHO, remove electron density most from the ortho and para positions, leaving the meta position relatively richer, so substitution occurs at meta and more slowly (meta directors). Halogens are a special case: deactivating but still o/p directing. Thus -COOH and -NO2 are meta directors, while -Cl, -CH3 and -OH are ortho/para directors.

5. Conversions

Toluene to o- and p-nitrobenzoic acid: toluene is nitrated with conc. HNO3/conc. H2SO4 to give a mixture of o-nitrotoluene and p-nitrotoluene, which is then oxidised with alkaline KMnO4 (or acidified K2Cr2O7) on heating, converting -CH3 into -COOH: O2N-C6H4-CH3 -> O2N-C6H4-COOH.

Benzene to m-nitrotoluene: the -NO2 (meta director) must be introduced before the -CH3 group. Step 1, nitration: C6H6 -> C6H5NO2. Step 2, Friedel-Crafts alkylation: C6H5NO2 + CH3Cl/anhydrous AlCl3 -> m-CH3-C6H4-NO2 + HCl, because the nitro group directs the incoming methyl group to the meta position.

Alkyl Halides and Amines

UNIT 6

Alkyl Halides

LESSON 41

Alkyl halides (haloalkanes) are organic compounds in which one or more hydrogen atoms of an alkane have been replaced by halogen atoms, general formula CnH2n+1X, for example C2H5Br. They are classified by the carbon atom carrying the halogen: in a primary (1 degree) alkyl halide the halogen-bearing carbon is attached to only one other carbon (composition RCH2X), for example 1-bromopropane; in a secondary (2 degree) alkyl halide it is attached to two other carbons (R2CHX); and in a tertiary (3 degree) alkyl halide it is attached to three other carbons (R3CX). CH3X is a methyl halide.

1. Orbital Structure of Methyl Iodide

In methyl iodide, CH3I, the carbon atom is sp3 hybridised: its one 2s and three 2p orbitals mix to form four sp3 hybrid orbitals arranged tetrahedrally with angles of about 109.5 degrees. Three sp3 orbitals overlap with the 1s orbitals of the three hydrogens to form three C-H sigma bonds. The fourth sp3 orbital overlaps with a p orbital of iodine to form the polar C-I sigma bond, which is longer and weaker than a C-H bond. The molecule is therefore tetrahedral, and the polarity of the C-I bond makes CH3I highly reactive in nucleophilic substitution.

2. Preparation from Alcohols

- **With HX:** R-OH + HX -> R-X + H2O, for example C2H5OH + HBr -> C2H5Br + H2O (ZnCl2 is used as catalyst with HCl). The reactivity order of the acids is HI > HBr > HCl, and tertiary alcohols react fastest.
- **With PX3:** 3R-OH + PX3 -> 3R-X + H3PO3, for example 3C2H5OH + PBr3 -> 3C2H5Br + H3PO3. This method gives good yields, especially for bromides and iodides prepared in situ from red phosphorus and the halogen.
- **With SOCl2:** R-OH + SOCl2 -> R-Cl + SO2 + HCl, for example C2H5OH + SOCl2 -> C2H5Cl + SO2 + HCl. The by-products are gaseous, so the halide is obtained pure; a little pyridine is often added to absorb the HCl formed.

3. Why Alkyl Halides are Insoluble in Water

Alkyl halides are polar molecules, but they cannot form hydrogen bonds with water because they have no hydrogen attached to a strongly electronegative atom. The energy released when halide molecules are surrounded by water is not enough to break the strong hydrogen bonds already present between water molecules. They therefore remain almost insoluble in water, although they dissolve readily in organic solvents of similar nature.

4. Nucleophiles

A nucleophile is an electron-pair donor, a species with a lone pair or negative charge that attacks an electron-deficient carbon centre and forms a new covalent bond. Four examples with their typical reagents are: hydroxide ion OH- (aqueous KOH or NaOH), cyanide ion CN- (KCN in alcohol), ammonia NH3 (alcoholic ammonia), and alkoxide ion RO- (sodium alkoxide, RONa). Halide ions such as I- (KI) are also common nucleophiles.

5. SN1 and SN2 Mechanisms

A nucleophilic substitution is a reaction in which a nucleophile replaces an atom or group (the leaving group) attached to a saturated carbon, for example R-X + OH- -> R-OH + X-. In the **SN1 mechanism** the reaction occurs in two steps: the halide slowly ionises to a planar carbocation and X-, and the nucleophile then attacks the carbocation rapidly from either side, giving a racemic product; the rate depends only on the concentration of the substrate, rate = k[RX], and tertiary halides react fastest. In the **SN2 mechanism** the nucleophile attacks the carbon from the back side at the same time as the leaving group departs, passing through a single transition state in which the carbon is partially bonded to both groups; because bond-making and bond-breaking are simultaneous and no carbocation intermediate is formed, the whole substitution is completed in one concerted step. The product shows complete inversion of configuration (Walden inversion), the rate depends on both [RX] and [Nu-], and the reaction is fastest when there is least steric hindrance: methyl iodide undergoes SN2 most easily, in the order methyl > primary > secondary.

6. Elimination Reactions: E1 and E2

Beta-elimination is the removal of a hydrogen atom from the beta-carbon (the carbon adjacent to the one bearing the leaving group) together with the leaving group itself, forming a double bond; dehydrohalogenation of alkyl halides with alcoholic KOH is the common example. Methyl bromide, CH3Br, has only one carbon and no beta-carbon at all, so it cannot form an alkene with alcoholic KOH. In the **E1 mechanism** the halide first ionises slowly to a carbocation, and a base then removes a beta-proton to give the alkene, so the rate depends only on [RX]. In the **E2 mechanism** a strong base removes the beta-proton at the same time as the halide leaves, in a single concerted step, so the rate depends on both [RX] and [base]. In both cases the more substituted alkene (Saytzeff product) usually predominates.

7. Substitution versus Elimination

In substitution a nucleophile replaces the halogen and the carbon remains saturated, for example C2HSBr + KOH (aq.) -> C2HSOH + KBr, while in elimination the halogen and a beta-hydrogen are removed and an unsaturated product is formed, C2HSBr + KOH (alc.) -> C2H4 + KBr + H2O. Substitution is favoured by aqueous alkali, lower temperature and a good nucleophile that is a weak base, whereas elimination is favoured by alcoholic alkali, higher temperature and a strong bulky base. Both reactions compete for the same substrate, substitution dominating for primary halides and elimination becoming more important for secondary and tertiary halides under basic conditions. Mechanistically, SN1 competes with E1 through a common carbocation, and SN2 competes with E2 as concerted pathways.

Grignard's Reagents (Organometallic Compounds)

LESSON 42

1. Organometallic Compounds and Preparation

Organometallic compounds are organic compounds containing a direct carbon-metal bond, for example CH3MgBr and (C2H5)4Pb. A Grignard reagent is prepared by adding an alkyl halide to dry magnesium turnings in dry ether: R-X + Mg -> R-MgX (dry ether), the ether and apparatus being kept completely free of moisture.

2. Why the Alkyl Part is Nucleophilic

In a Grignard reagent, RMgX, magnesium is much less electronegative than carbon, so the C-Mg bond is polarised as C delta- and Mg delta+. The alkyl group therefore carries a partial negative charge and a pair of electrons it can donate, behaving like a carbanion. It attacks electron-deficient (electrophilic) centres such as the carbonyl carbon, so the alkyl part acts as a nucleophile and a strong base.

3. Reactions of Grignard Reagents

- **With water:** RMgX + H2O -> R-H + Mg(OH)X. Water is the standard reagent used to hydrolyse a Grignard reagent: the alkyl group takes a proton from water and the corresponding alkane is formed. The alkane formed always has the same carbon skeleton as the Grignard alkyl group, so methyl magnesium iodide itself gives methane, CH3MgI + H2O -> CH4 + Mg(OH)I; propane is obtained by the same method from propyl magnesium halide, C3H7MgX + H2O -> C3H8.
- **With carbon dioxide:** RMgX + CO2 -> R-COOH after hydrolysis, a carboxylic acid with one extra carbon. Ethyl magnesium bromide gives propanoic acid: C2HSMgBr + CO2 -> C2HSCOOH.
- **With an ester:** the reagent first displaces the alkoxy group of the ester to give a ketone, RCOOR + RMgX -> R-CO-R; with a second equivalent of Grignard reagent the ketone reacts further to give a tertiary alcohol.
- **With amines** (or any compound with an acidic hydrogen): the reagent is simply protonated to the alkane, RMgX + H2N-R -> R-H + BrMg-NH-R.

4. Conversion: Methyl Magnesium Bromide into Acetone

Step 1: methyl magnesium bromide is treated with carbon dioxide and the adduct is hydrolysed to give ethanoic acid: CH3MgBr + CO2 -> CH3COOH (after hydrolysis). Step 2: the acid is neutralised with Ca(OH)2 to give calcium acetate, (CH3COO)2Ca. Step 3: dry distillation of calcium acetate gives acetone: (CH3COO)2Ca -> CH3-CO-CH3 + CaCO3 on strong heating.

Amines

LESSON 43

1. Structure and Basic Character

Amines act as bases because the nitrogen atom has a lone pair of electrons which it donates to a proton, forming an alkylammonium ion: R-NH2 + H+ -> R-NH3+. Accepting a proton is the behaviour of a base. In a primary amine the nitrogen is bonded to three atoms (one C and two H) and carries one lone pair, giving a tetrahedral pyramidal structure similar to ammonia, with a bond angle of about 107 degrees, slightly less than the tetrahedral angle because the lone pair repels the bond pairs. In water: R-NH2 + H2O <-> R-NH3+ + OH-. Basic strength is expressed by Kb or by the pKa of the conjugate acid.

2. Basicity of Amines

Alkyl groups are electron-donating (+I effect), so they push electron density onto the nitrogen and make its lone pair more available for accepting a proton; they also stabilise the alkylammonium ion formed. A secondary amine has two alkyl groups and a tertiary amine has three, so the effect is greater than in a primary amine with only one alkyl group, which is why secondary and tertiary amines are more alkaline than primary amines. In aqueous solution the order is modified by solvation and steric effects, but the inductive effect explains why substituted amines are stronger bases than ammonia and, in the gas phase, tertiary > secondary > primary. Aromatic amines such as aniline are much weaker bases, because the nitrogen lone pair is delocalised into the benzene ring and is less available for protonation.

3. Preparations

From nitriles: nitriles (alkyl cyanides) are reduced with hydrogen in the presence of a nickel catalyst, or with LiAlH4 or sodium and alcohol, to give primary amines having one carbon more than the original alkyl halide used to make the nitrile: R-C triple bond N + 4[H] -> R-CH2-NH2, for example CH3-CN + 4[H] -> CH3-CH2-NH2 (ethylamine). This is a useful method because the chain is lengthened by one carbon atom.

From alkyl halides (ammonolysis): ethyl chloride is heated with an excess of alcoholic ammonia under pressure, when the chlorine is replaced by the -NH2 group: C2H5Cl + NH3 (alc., excess, heat) -> C2H5NH2 + HCl, the amine being obtained as its salt and liberated by alkali. Excess ammonia is used to reduce further alkylation. On further alkylation a primary amine gives a secondary amine and then a tertiary amine: R-NH2 + R-X -> R2NH + HX, and R2NH + R-X -> R3N + HX; with excess RX the tertiary amine gives the quaternary salt [R4N]+ X-.

4. Reaction with Nitrous Acid

A primary aliphatic (alkyl) amine reacts with nitrous acid (NaNO2 + HCl) to form an unstable diazonium salt which immediately decomposes, giving the corresponding alcohol with evolution of nitrogen: R-NH2 + HNO2 -> R-OH + N2 + H2O. Stable diazonium salts are formed only by aromatic primary amines at low temperature.

5. Conversions

Ethyl chloride into ethyl alcohol: C2H5Cl + KOH (aq.) -> C2H5OH + KCl, by boiling with aqueous potassium hydroxide (or moist silver oxide); alcoholic KOH must be avoided because it gives ethene instead.

Ethylamine into imine: ethylamine is dehydrogenated (oxidised) to the imine ethanimine, in which the -NH2 group becomes a C=N double bond: CH3-CH2-NH2 -> CH3-CH=NH + H2. Imine formation by condensation of an amine with an aldehyde or ketone with loss of water gives substituted imines (Schiff bases) in the same family.

Alcohols, Phenols and Ethers

UNIT 7

Alcohols

LESSON 44

Alcohols are organic compounds containing one or more hydroxyl (-OH) groups attached to an aliphatic carbon atom, general formula R-OH or CnH2n+1OH. They are classified in two ways: by the number of -OH groups, as monohydric (one -OH, e.g. ethanol), dihydric (two -OH, e.g. ethylene glycol) and trihydric/polyhydric (three or more, e.g. glycerol); and by the carbon bearing the -OH group, as primary (1 degree), secondary (2 degree) and tertiary (3 degree) alcohols.

1. Physical Properties

The -OH group of an alcohol can form hydrogen bonds with water molecules, and this attraction compensates for breaking the hydrogen bonds in water itself, so lower alcohols such as methanol and ethanol are miscible with water in all proportions. Solubility decreases as the non-polar alkyl chain becomes longer, because the hydrocarbon part opposes dissolution. Ethanol is a liquid at room temperature (b.p. 78 C) because its molecules are held together by strong intermolecular hydrogen bonds and are associated into larger clusters, whereas ethyl chloride has no hydrogen bonding and only weak van der Waals forces, so it escapes easily as a gas (b.p. about 12 C). Boiling point increases with molecular mass and with chain straightness, because a straight chain has a larger surface area and stronger van der Waals forces: n-pentyl alcohol (pentan-1-ol, b.p. about 138 C) boils higher than its branched isomers isopentyl and neopentyl alcohol, and much higher than ethyl alcohol.

2. Preparation from Grignard Reagents

- **Primary alcohol:** RMgX + HCHO -> R-CH2-OMgX -> R-CH2OH (with H2O), from formaldehyde, the alcohol having one carbon more than R.
- **Secondary alcohol:** RMgX + R-CHO -> R2CH-OH after hydrolysis, from any other aldehyde.
- **Tertiary alcohol:** RMgX + R-CO-R -> R3C-OH after hydrolysis, from a ketone, the carbonyl carbon ending up attached to three carbon groups and one -OH group.

3. Important Reactions

- **With PCl3:** 3C2H5OH + PCl3 -> 3C2H5Cl + H3PO3. Phosphorus trichloride replaces the -OH group by chlorine, giving ethyl chloride, a gas used as a local anaesthetic spray.
- **Dehydration:** at 170 C with concentrated H2SO4 ethanol gives ethene, C2H5OH -> CH2=CH2 + H2O (intramolecular dehydration); at about 140 C two molecules lose one molecule of water between them (intermolecular dehydration) giving diethyl ether: 2C2H5OH -> C2H5-O-C2H5 + H2O (conc. H2SO4, 140 C). Excess ethanol and the lower temperature favour ether formation.
- **Hydration of ethene (industrial ethanol):** CH2=CH2 + H2SO4 -> CH3CH2OSO3H, and CH3CH2OSO3H + H2O -> C2H5OH + H2SO4 (boil). Ethene is absorbed in hot concentrated sulphuric acid to form ethyl hydrogen sulphate, which on hydrolysis with boiling water gives ethanol.
- **Oxidation:** with acidified potassium dichromate, ethanol is oxidised first to ethanal and then to ethanoic (acetic) acid, the orange dichromate turning green: CH3CH2OH + 2[O] -> CH3COOH + H2O. If the ethanal is distilled off as it forms, it can be isolated instead. A secondary alcohol is first oxidised to a ketone, R2CH-OH -> R2C=O, which is the intermediate product, and only under vigorous conditions is the ketone further cleaved to carboxylic acids with fewer carbons (for example propan-2-ol -> propanone -> ethanoic acid).
- **Reduction of acetic acid:** CH3COOH -> CH3CH2OH with LiAlH4 in dry ether followed by H2O work-up; the -COOH group is converted into -CH2OH.
- **Ethyl bromide to ethanol:** C2H5Br + KOH (aq.) -> C2H5OH + KBr, by boiling with aqueous KOH or moist Ag2O; alcoholic KOH would give ethene by elimination.

4. Cleavage of 1,2-Diols

Oxidative cleavage of a 1,2-diol (vicinal diol) occurs in the presence of periodic acid, HIO4, which specifically cleaves the C-C bond, each -CH(OH)- group being converted into a carbonyl group. For example, ethylene glycol gives two molecules of formaldehyde: HOCH2-CH2OH + HIO4 -> 2HCHO + HIO3 + H2O.

5. Lucas Test

Lucas reagent is a mixture of anhydrous ZnCl2 and concentrated HCl, which converts alcohols into insoluble alkyl chlorides that make the solution turbid. A tertiary alcohol reacts immediately and the mixture becomes turbid at once. A secondary alcohol becomes turbid within about five minutes. A primary alcohol does not react at room temperature and the mixture stays clear, reacting only on heating. The test works because tertiary carbocations form most easily.

6. Identification Tests for an Alcohol

(1) Sodium metal test: an alcohol gives brisk effervescence of hydrogen with a small piece of sodium, 2R-OH + 2Na -> 2R-ONa + H2. (2) Ester test: warmed with acetic acid and a few drops of concentrated H2SO4 it gives a sweet, fruity smell of the ester formed. Ethanol also gives the iodoform test, a yellow precipitate of CHI3 with I2/NaOH.

7. Commercial Applications of Alcohols

Ethanol is used as a fuel and fuel additive, as a solvent for paints, varnishes, perfumes and medicines, in alcoholic drinks and as an antiseptic; methanol is used to make formaldehyde, plastics and as an industrial solvent, and ethylene glycol is used as antifreeze.

Phenols

LESSON 45

Phenol, C6H5OH, is the aromatic compound in which a hydroxyl group is attached directly to a benzene ring; it is also called carbolic acid because it is weakly acidic. All carbon atoms of the benzene ring in phenol, including the one bearing the -OH group, are sp2 hybridised and part of the flat aromatic ring, which allows the oxygen lone pair to conjugate with the ring pi system. Pyrogallol (benzene-1,2,3-triol) is a trihydric phenol with three -OH groups on the ring; resorcinol and catechol are dihydric and cresol is monohydric.

1. Preparation of Phenol

- **From chlorobenzene (Dow process):** chlorobenzene is fused with aqueous NaOH at about 350 C and 150 atmospheres pressure to give sodium phenoxide, which on acidification gives phenol: C6H5Cl + 2NaOH -> C6H5ONa + NaCl + H2O, then C6H5ONa + HCl -> C6H5OH + NaCl.
- **From sodium benzene sulphonate:** the salt is fused with NaOH at about 300 C to give sodium phenoxide and sodium sulphite: C6H5SO3Na + 2NaOH -> C6H5ONa + Na2SO3 + H2O, and the phenoxide is then acidified to phenol.
- **From diazonium salt:** a benzene diazonium salt solution, prepared by diazotising aniline at 0-5 C, on warming with water hydrolyses to phenol with evolution of nitrogen: C6H5N2+ Cl- + H2O -> C6H5OH + N2 + HCl (warm).

2. Alcohol versus Phenol

- **Solubility:** lower alcohols such as ethanol are completely miscible with water because the small alkyl group allows extensive hydrogen bonding with water. Phenol is only sparingly soluble in cold water (about 8 g per 100 mL) because its large non-polar benzene ring opposes dissolution, although it becomes fully miscible above about 66 C.
- **Boiling point:** phenol boils at about 182 C, much higher than ethanol (78 C), partly because of its larger molecular mass and strong hydrogen bonding combined with ring interactions. Alcohols also boil high for their mass due to hydrogen bonding, increasing with chain length (methanol 65 C, ethanol 78 C, propan-1-ol 97 C).
- **Acidic character:** phenol is measurably acidic (pKa about 10): it turns blue litmus red, reacts with NaOH to form sodium phenoxide and water, and its phenoxide ion is stabilised by resonance delocalisation into the ring. Alcohols are essentially neutral (pKa about 16-18), do not affect litmus and do not react with NaOH, because the alkoxide ion gets no such resonance stabilisation and the alkyl group pushes electrons onto oxygen. Phenol is therefore a much stronger acid than an alcohol, though weaker than carboxylic acids and carbonic acid.

3. Reactions of Phenol

- **With hot concentrated nitric acid:** phenol is both nitrated and oxidised, the main isolated product being 2,4,6-trinitrophenol, picric acid, a yellow crystalline explosive: C6H5OH + 3HNO3 (conc., heat) -> C6H2(NO2)3-OH + 3H2O. In practice picric acid is usually made via sulphonation first to reduce oxidation losses.
- **With concentrated sulphuric acid at 100 C:** phenol is sulphonated mainly at the para position, giving p-phenol sulphonic acid (4-hydroxybenzene sulphonic acid): C6H5OH + H2SO4 -> p-HO-C6H4-SO3H + H2O. The ortho product predominates at room temperature.
- **With bromine water:** phenol instantly decolourises bromine water and gives a white precipitate of

2,4,6-tribromophenol, because the -OH group strongly activates the ring at both ortho positions and the para position: C6H5OH + 3Br2 -> C6H2Br3-OH + 3HBr. This reaction is a confirmatory test for phenol.

- **With sodium metal:** 2C6H5OH + 2Na -> 2C6H5ONa (sodium phenoxide) + H2, less vigorously than sodium with water or with carboxylic acids, in keeping with the weak acidity of phenol.
- **Oxidation to benzoquinone:** C6H5OH + [O] -> O=C6H4=O (p-benzoquinone) + H2O, with Na2Cr2O7/H2SO4; a second carbonyl appears at the para position of the ring.

4. Identification Tests for Phenol

(1) Ferric chloride test: phenol gives a violet (purple) colouration with neutral FeCl3 solution. (2) Bromine water test: phenol decolourises bromine water and gives a white precipitate of 2,4,6-tribromophenol. It also turns blue litmus slightly red, showing its weak acidity.

5. Commercial Applications of Phenol

Phenol is used to make Bakelite and other phenol-formaldehyde resins, nylon intermediates, dyes, antiseptics such as TCP and Dettol-type products, aspirin and picric acid.

Ethers

LESSON 46

1. Physical Properties

Alcohol molecules are associated by strong intermolecular hydrogen bonding through their -OH groups, so much heat is needed to separate them. Ether molecules have no hydrogen attached to oxygen and cannot form hydrogen bonds among themselves; only weak dipole forces hold them together. Ethers therefore boil at much lower temperatures than alcohols of comparable molecular mass, and they cannot form hydrogen bonds with water through an -OH group, so compounds like ethyl chloride and the higher ethers are practically insoluble in water.

2. Oxonium Ion Formation

An oxonium ion is a positively charged oxygen species in which oxygen carries three bonds and one lone pair, for example H3O+ and R2OH+. An ether oxygen has two lone pairs; in strong acid it accepts a proton on one lone pair to form a dialkyloxonium ion: R-O-R + H+ -> [R-O(H)-R]+, for example (C2H5)2O + HCl -> [(C2H5)2OH]+ Cl-. Formation of oxonium salts in concentrated acids explains why ethers dissolve in concentrated H2SO4 and how ether linkages are cleaved by hot concentrated HX.

3. Ethers as Solvents

Ethers are good solvents because they dissolve a wide range of organic compounds, both polar and non-polar, yet they are chemically quite unreactive towards acids, bases, oxidising and reducing agents under normal conditions. They do not react with Grignard reagents and organometallic compounds, which is why reactions like the preparation of RMgX are carried out in dry ether. Their low boiling points also make them easy to remove after the reaction.

4. Diethyl Ether

Diethyl ether is formed by heating ethanol with concentrated H2SO4 at about 140 C: 2C2H5OH -> C2H5-O-C2H5 + H2O. It was the first widely used general anaesthetic and produces unconsciousness when its vapour is inhaled; it is highly volatile and inflammable, which is why safer anaesthetics have largely replaced it in modern surgery. Ethers in general serve as solvents for oils, fats and Grignard reactions and as starting materials in industry.

Carbonyl Compounds I - Aldehydes and Ketones

UNIT 8

Physical Properties

LESSON 47

1. Boiling Points

The boiling points of aldehydes and ketones are lower than those of alcohols. Alcohols have -OH groups and form strong intermolecular hydrogen bonds, while aldehydes and ketones have no hydrogen attached to oxygen, so they are held only by weaker dipole-dipole forces and need less heat to boil.

2. Solubility in Water

Formaldehyde is the most soluble carbonyl compound in water: it is the smallest aldehyde, its carbonyl oxygen hydrogen-bonds strongly with water, and there is no large non-polar hydrocarbon part to oppose dissolving. As the size of the alkyl or aryl group increases, as in acetaldehyde, benzaldehyde and acetophenone, the molecule becomes less polar and solubility in water decreases, so formaldehyde is highly soluble compared with the other aldehydes.

Structure

LESSON 48

Aldehydes are organic compounds containing the -CHO group, in which the carbonyl carbon is attached to at least one hydrogen, general formula R-CHO, while ketones contain the C=O group between two carbon groups, general formula R-CO-R. Acetophenone, C6H5-CO-CH3, is a member of the ketone family containing one alkyl group and one aryl group on the carbonyl carbon; a hydrogen on the carbonyl carbon would make it an aldehyde, not a ketone.

1. Hybridisation of the Carbonyl Carbon

In both aldehydes and ketones the carbonyl carbon is sp2 hybridised: it forms three sigma bonds (to oxygen and to two other atoms) in a trigonal planar arrangement with angles of about 120 degrees, and the fourth valence forms a pi bond with oxygen using the unhybridised p-orbital. The C=O bond is polar because oxygen is more electronegative, so carbon carries a partial positive charge and oxygen a partial negative charge, which explains their nucleophilic addition reactions.

Preparations of Aldehydes and Ketones

LESSON 49

1. Formaldehyde by Ozonolysis

Formaldehyde is obtained by the ozonolysis of ethene. Ethene reacts with ozone to form an unstable ozonide, which on hydrolysis with zinc and water breaks at the double bond: CH2=CH2 + O3 -> ozonide -> (Zn/H2O) 2HCHO + H2O. Each CH2 end of the double bond becomes one molecule of formaldehyde, and zinc prevents its further oxidation to formic acid.

2. Acetone by Hydration of Propyne

Water adds to propyne CH3-C triple bond CH according to Markownikov rule in the presence of H2SO4/HgSO4, giving an unstable enol CH3-C(OH)=CH2. This enol tautomerises to the more stable ketone, acetone CH3-CO-CH3.

3. Acetaldehyde by Hydration of Ethyne

HC triple bond CH + H2O -> CH3CHO in the presence of H2SO4/HgSO4. The major product is acetaldehyde (ethanal), formed through the unstable vinyl alcohol enol.

Reactivity

LESSON 50

1. Order of Reactivity

Formaldehyde is the most reactive carbonyl compound towards nucleophilic addition. It has two small hydrogen atoms on the carbonyl carbon, so there is almost no steric hindrance and no electron-donating alkyl group to reduce the positive charge on carbon; a nucleophile can therefore attack its carbonyl carbon most easily. In ketones two bulky alkyl groups donate electrons towards the carbonyl carbon, reducing its positive charge, and they also block the approach of the nucleophile sterically. The order of reactivity is: formaldehyde > other aldehydes > ketones, diethyl ketone and acetophenone being the least reactive of the common examples.

2. Factors Influencing Reactivity

Reactivity depends mainly on the charge on the carbonyl carbon and on steric hindrance around it.
Electron-donating alkyl groups reduce the partial positive charge on carbon and make attack harder, and bulky groups near the carbonyl carbon block the approach of the nucleophile, so larger alkyl or aryl groups lower the reactivity, as in acetophenone. Strong nucleophiles and base catalysis favour addition with weak carbonyl compounds, while acid catalysis, by protonating the oxygen, increases the positive charge on carbon and helps weak nucleophiles to attack.

3. Acid-Catalysed and Base-Catalysed Addition

In base-catalysed addition, used with strong nucleophiles, the nucleophile Nu- attacks the carbonyl carbon first, forming an alkoxide intermediate, which then takes a proton from water to give the addition product, e.g. HCHO + HCN -> HCH(OH)CN in basic medium. In acid-catalysed addition, used with weak nucleophiles like water, alcohols and ammonia, the acid first protonates the carbonyl oxygen, which increases the positive charge on carbon and makes it more attractive for the weak nucleophile. The nucleophile then attacks the carbon, and loss of a proton gives the final product such as a hemiacetal. Thus acid activates the carbonyl compound, while base activates or supplies a strong nucleophile.

Reactions of Aldehydes and Ketones

LESSON 51

1. Reactions of Acetaldehyde

- **With chromic acid:** CH3CHO + [O] -> CH3COOH (H2CrO4); acetaldehyde is oxidised to acetic acid.
- **With LiAlH4:** CH3CHO + 2[H] -> CH3CH2OH; it is reduced to ethanol, a primary alcohol.
- **With Zn-Hg and conc. HCl (Clemmensen reduction):** CH3CHO + 4[H] -> CH3CH3 + H2O; the C=O is reduced to CH2 and ethane is formed.
- **With NaBH4:** CH3CHO + 2[H] -> CH3CH2OH; the major product is ethanol.

2. Reduction Reactions

Clemmensen reduction is the conversion of aldehydes and ketones into alkanes by zinc amalgam Zn-Hg and concentrated HCl under strongly acidic conditions; the carbonyl group C=O is completely reduced to a CH2 group, for example CH3CHO -> CH3CH3. Wolf-Kishner reduction also gives alkanes but uses hydrazine NH2NH2 and strong base KOH instead. With LiAlH4, ketones are reduced to secondary alcohols: the hydride adds to the carbonyl carbon and R-CO-R gives R-CHOH-R, in which the -OH is attached to a carbon bonded to two other carbons.

3. Oxidation: Aldehydes versus Ketones

An aldehyde has a hydrogen on the carbonyl carbon which is easily removed during oxidation to a carboxylic acid, but a ketone has no such hydrogen and oxidation requires breaking a strong C-C bond, so aldehydes oxidise much faster and even with mild oxidising agents.
Aldehydes are oxidised by Tollens reagent, Fehling solution and Benedict solution to a carboxylic acid with the same number of carbon atoms, e.g. CH3CHO -> CH3COOH.
Ketones do not react with these mild reagents and are oxidised only by strong oxidising agents like acidified K2Cr2O7 with C-C bond cleavage, giving carboxylic acids with fewer carbons: CH3COCH3 + 4[O] -> CH3COOH + HCOOH (formic acid further oxidises to CO2 and H2O), the major carboxylic acid product being acetic acid.

4. Tests that Distinguish Aldehydes from Ketones

Tollens reagent is ammoniacal silver nitrate [Ag(NH3)2]OH. On warming with an aldehyde it is reduced to metallic silver, which deposits as a bright silver mirror on the test tube: RCHO + 2[Ag(NH3)2]OH -> RCOOH + 2Ag + 4NH3 + H2O; ketones give no mirror. Formaldehyde, HCHO, gives the silver mirror test with Tollens reagent. **Fehling solution** is an alkaline solution of copper sulphate complexed with tartrate. On heating with an aldehyde the blue Cu2+ is reduced to a red precipitate of cuprous oxide Cu2O: RCHO + 2Cu(OH)2 + NaOH -> RCOONa + Cu2O + 3H2O; ketones give no red precipitate.

5. Nucleophilic Addition Reactions of Formaldehyde

- **With hydrogen cyanide:** HCHO + HCN -> CH2(OH)CN, the cyanohydrin hydroxyacetonitrile (glycolonitrile), in the presence of a base catalyst.
- **With a primary alcohol:** HCHO + 2ROH -> CH2(OR)2 + H2O under acid catalysis, first a hemiacetal and then an acetal.
- **With methyl magnesium bromide:** HCHO + CH3MgBr -> CH3CH2OMgBr -> (H2O/H+) CH3CH2OH, ethanol, a primary alcohol with one extra carbon.
- **With ammonia:** formaldehyde does not give a simple imine; six molecules condense with four of ammonia to form hexamethylenetetramine (urotropine): 6HCHO + 4NH3 -> (CH2)6N4 + 6H2O.

6. Four Differences between Aldehydes and Ketones

- An aldehyde has the carbonyl group at the end of the chain with at least one hydrogen attached (R-CHO), while a ketone has it in the middle, attached to two carbon groups (R-CO-R).
- Aldehydes are easily oxidised to carboxylic acids by mild oxidising agents, while ketones resist mild oxidising agents and need strong ones.
- Aldehydes give the Tollens silver mirror test and the Fehling red precipitate test, while ketones give neither test.
- Aldehydes are more reactive towards nucleophilic addition than ketones because they have less steric hindrance and a more positively charged carbonyl carbon.

Carbonyl Compounds II - Carboxylic Acid and Functional Derivatives

UNIT 9

Physical Properties

LESSON 52

1. Boiling Point

Carboxylic acids have unusually high boiling points, higher than alcohols of similar mass, because their molecules associate into cyclic dimers through two strong hydrogen bonds. The acids are more polar than alcohols and form very strong hydrogen bonds between two molecules, so a large amount of heat energy is needed to break these bonds and boil the acid. Acetic acid (boiling point 118 C) therefore boils higher than ethanol (78 C), acetaldehyde (20 C) and ethyl chloride (12 C). Alcohols form weaker, single hydrogen-bonded associations, hence acids of comparable mass boil higher than alcohols.

2. Solubility

The lower carboxylic acids up to four carbons are miscible or highly soluble in water because the -COOH group forms hydrogen bonds with water molecules. As the alkyl chain grows, the non-polar hydrocarbon part dominates, hydrogen bonding becomes less effective, and solubility falls rapidly, so higher fatty acids are practically insoluble in water though soluble in organic solvents.

Structure

LESSON 53

A carboxylic acid contains the carboxyl group -COOH, which is a combination of a carbonyl group C=O and a hydroxyl group -OH on the same carbon. This carbon is sp2 hybridised and forms three sigma bonds - to R, to =O oxygen and to -OH oxygen - which arrange themselves in one plane at about 120 degrees to each other, so the -COOH structure is trigonal planar. The lone pairs on oxygen conjugate with the carbonyl pi bond, giving resonance in the group. Because of the polar C=O and the -OH, two molecules join by two hydrogen bonds into a cyclic dimer, which explains the high boiling points and association of carboxylic acids.

Acidity

LESSON 54

Carboxylic acids turn blue litmus red, react with metals, carbonates and alkalis, and lose the hydroxyl proton in water: RCOOH + H2O -> RCOO- + H3O+. They are stronger acids than alcohols and phenols because the carboxylate ion RCOO- is stabilised by resonance, the negative charge being shared equally by both oxygens, so the proton is lost easily. They are much stronger acids than ethanol, but weak acids compared with mineral acids like HCl, HNO3 and H2SO4, because their ionisation is only partial (acetic acid is only a few percent ionised) - the equilibrium lies to the left, whereas mineral acids ionise almost completely in water.

Preparations of Carboxylic Acids and their Derivatives

LESSON 55

1. Preparation of Carboxylic Acids

- **Carbonation of Grignard reagent:** a Grignard reagent RMgX is treated with dry carbon dioxide (dry ice); the reagent adds to CO2 and the adduct on hydrolysis with dilute acid gives a carboxylic acid with one carbon more than R: RMgX + CO2 -> RCOOMgX -> (H3O+) RCOOH. For example, ethyl magnesium bromide gives propanoic acid: C2H5MgBr + CO2 -> C2H5COOMgBr -> (H3O+) C2H5COOH.
- **Hydrolysis of alkyl nitriles:** an alkyl nitrile RCN is heated with dilute acid or alkali; it hydrolyses first to an amide and then to the acid: RCN + 2H2O -> (H+/heat) RCOOH + NH3.
- **Oxidation of primary alcohols:** a primary alcohol is oxidised with a strong oxidising agent like acidified K2Cr2O7 or KMnO4, passing through the aldehyde stage to the acid with the same number of carbons: RCH2OH -> [O] RCHO -> [O] RCOOH; for example ethanol gives acetic acid.
- **Oxidation of alkyl benzenes:** alkyl benzenes with a benzylic carbon are oxidised by strong oxidising agents like alkaline KMnO4 or acidified K2Cr2O7 to benzoic acid, whatever the length of the side chain:

C6H5CH2CH3 -> [O] C6H5COOH, and toluene likewise gives benzoic acid: C6H5CH3 -> [O] C6H5COOH.

2. Formation of the Functional Derivatives

All functional derivatives of carboxylic acids are formed by replacing the -OH group of -COOH by another group: replacement by Cl gives an acyl halide RCOCl, by NH2 gives an amide RCONH2, by OR gives an ester and by OCOR gives an anhydride. Acyl halides are prepared using PCl3, PCl5 or thionyl chloride SOCl2, for example RCOOH + SOCl2 -> RCOCl + SO2 + HCl. Hydrochloric acid HCl alone cannot do this replacement, because it is not a chlorinating agent for the hydroxyl group.

Reactions of Carboxylic Acids and their Derivatives

LESSON 56

1. The Four Derivatives from Acetic Acid

- **Acetyl chloride (acyl halide):** CH3COOH + SOCl2 -> CH3COCl + SO2 + HCl
- **Acetic anhydride (acid anhydride):** 2CH3COOH -> (P2O5/heat) (CH3CO)2O + H2O. On strong heating, or with a dehydrating agent like P2O5, two molecules of acetic acid lose one molecule of water and join through an oxygen atom.
- **Ethyl acetate (ester):** CH3COOH + C2H5OH -> (conc. H2SO4) CH3COOC2H5 + H2O. This is esterification; the ester formed has a sweet fruity smell.
- **Acetamide (amide):** CH3COOH + NH3 -> CH3COONH4 -> (heat) CH3CONH2 + H2O. The acid first forms an ammonium salt, which on heating loses water to give the amide.

2. Reactions of the Derivatives

An ester is converted back into a carboxylic acid by hydrolysis: heating with dilute acid or alkali; with alkali the salt is formed, which on acidification gives the acid: CH3COOC2H5 + H2O -> (H+/heat) CH3COOH + C2H5OH. Esters give fruits their pleasant odours: the characteristic fruity smell and flavour of pineapple is due to the ester ethyl butanoate, and many fruits get their odours from such low molar mass esters.

3. Formation of Formic Acid by Oxidation

Formaldehyde is fully oxidised by a mixture of potassium dichromate and sulphuric acid, first to formic acid and then to carbon dioxide and water: HCHO -> (K2Cr2O7/H2SO4) HCOOH -> CO2 + H2O, and the orange dichromate turns green.

4. Natural Sources of Carboxylic Acids

- **Formic acid (methanoic acid, HCOOH):** occurs in the sting of bees, from where it enters the skin and causes pain and irritation (according to the book), and is also associated with ant venom.
- **Acetic acid:** present in vinegar, formed by the fermentation of ethanol in air.
- **Valeric acid (pentanoic acid):** occurs in the valerian plant root, from which it takes its name.
- **Caproic acid (hexanoic acid):** occurs in goat milk and goat fat (capra means goat) and in small amounts in other animal fats. Counting the carboxyl carbon, its formula is C5H11COOH, that is C6H12O2 in total.

5. Commercial Applications

Carboxylic acids are used as preservatives, flavouring agents and acidity regulators in the food industry, for example acetic acid as vinegar and benzoic acid as a preservative. They are starting materials for making soaps (salts of long-chain fatty acids), detergents, perfumes and esters used as solvents and flavours. They are also used in making medicines such as aspirin, in the manufacture of polymers like nylon and polyester, and in the rubber, textile dyeing and leather industries.

Biochemistry

UNIT 10

Carbohydrates

LESSON 57

Carbohydrates are polyhydroxy aldehydes and ketones, or substances that yield such compounds on hydrolysis; sugars such as glucose, fructose and sucrose belong to this class. They are the main source of energy for the body, with sources like rice, potato, wheat and barley.

1. Classification on the Basis of Structure

- **Monosaccharides:** a single sugar unit, cannot be hydrolysed further, e.g. glucose, fructose and galactose.
- **Disaccharides:** two units, e.g. sucrose, lactose and maltose. Sucrose (table sugar) is made of glucose and fructose.
- **Oligosaccharides:** 3 to 10 units, e.g. kestose and melezitose.
- **Polysaccharides:** more than 10 units, e.g. starch, cellulose and glycogen. Starch is made of very many glucose units joined together and is used as energy storage in plants.

2. Role of the Common Sugars in Health

- **Glucose:** the immediate source of energy for cells and the normal sugar of blood; its level is controlled by insulin, and its failure leads to diabetes while its shortage causes weakness.
- **Fructose:** the sweetest natural sugar, taken from sweet fruits such as grapes, apples, mangoes and dates and from honey; it is metabolised in the liver and provides energy, but excess intake burdens the liver. Table sugar sucrose also gives fructose on hydrolysis.
- **Sucrose:** the common table sugar from sugarcane and beet, hydrolysed to glucose and fructose and supplying quick energy, though excess causes obesity and tooth decay.
- **Lactose:** the milk sugar, obtained only from milk and milk products such as curd and cheese; it provides energy to infants and helps absorption of calcium, but people lacking the enzyme lactase cannot digest it properly (lactose intolerance). On digestion it is hydrolysed to glucose and galactose.

3. Dietary Importance

Carbohydrates are the main and quickest source of energy for our tissues and organs; glucose is the immediate fuel, especially for the brain, and 1 g gives about 4 kcal. They spare proteins from being used as fuel, store extra energy as glycogen in liver and muscles and as starch-rich foods, and provide dietary fibre (cellulose) that aids digestion and bowel movement.

Proteins

LESSON 58

Proteins are natural condensation polymers whose building blocks are alpha amino acids. According to the book, there are 22 alpha amino acids in total that take part in making proteins in living organisms; some can be made by the body itself while the essential ones must be supplied through diet. In a protein, the -COOH group of one amino acid joins with the -NH2 group of the next amino acid with loss of a water molecule, forming a -CO-NH- bond called the peptide linkage, and long chains of such linkages make the protein. On hydrolysis proteins give back these amino acids.

1. Classification on the Basis of Function

- **Structural proteins** (collagen, keratin) that build tissues.
- **Enzymes** (amylase, pepsin) that catalyse body reactions.
- **Hormones** (insulin) that regulate processes.
- **Transport proteins** (haemoglobin) that carry substances.
- **Protective proteins** (antibodies) that fight infection.
- **Contractile proteins** (actin, myosin) that cause muscle movement.
- **Storage proteins** (casein of milk) that store amino acids and nutrients.

2. Structure of Proteins

The **primary structure** is the specific sequence in which amino acids are joined by peptide bonds in the polypeptide chain; it decides all higher structures of the protein. The **secondary structure** is the regular folding of this chain into an alpha-helix or a beta-pleated sheet, held by hydrogen bonds between C=O and N-H groups of nearby peptide bonds. The **tertiary structure** is the overall three-dimensional folding of the whole chain upon itself, stabilised by hydrogen bonds, ionic bonds, disulphide bonds and hydrophobic interactions, and it is this final shape that gives the protein its biological activity.

3. Physical Properties of Proteins

Proteins are colourless, mostly tasteless, high molar mass macromolecular solids; some are crystalline while most are amorphous. Some proteins like albumin of egg are soluble in water forming colloidal solutions, while fibrous proteins like keratin are insoluble in water and common solvents. They are denatured (their structure is destroyed) by heat, strong acids, alkalis, alcohol and heavy metal salts, as seen when egg white sets on heating. Proteins also show optical activity, can be precipitated from solution, and on strong heating they char and give a characteristic burnt-hair smell.

4. Essential Functions in the Body

Proteins are the body-building material: they build and repair muscles, skin and other tissues and are essential for growth. Many proteins act as enzymes and hormones that control and speed up the chemical reactions and processes of the body, such as insulin and digestive enzymes. Proteins like antibodies defend the body against disease, haemoglobin carries oxygen, and in need proteins can also be oxidised to provide energy.

Lipids

LESSON 59

Lipids are water-insoluble, heterogeneous organic compounds which are chemically esters of fatty acids and alcohols, soluble in organic solvents like ether and chloroform. They are classified into simple lipids (fats, oils and waxes), compound lipids (phospholipids and glycolipids) and derived lipids (steroids, fatty acids, glycerol). Cell membranes are built mainly of a double layer of phospholipids, whose hydrophilic phosphate heads face the water on both sides and whose hydrophobic fatty acid tails face inwards; triglycerides are storage fats, while glycolipids and steroids are present in membranes only in smaller amounts.

1. Properties and Biological Significance

Their properties include insolubility in water, greasy feel, lower density than water, hydrolysis by alkali (saponification) and becoming rancid on oxidation. The three main functions of lipids are: first, they act as a concentrated store of energy and provide about twice as much energy per gram as carbohydrates; second, they form the structural material of cell membranes, especially phospholipids, and protect delicate internal organs as a cushion; third, they act as thermal insulators under the skin, prevent water loss, and serve as solvents and carriers for the fat-soluble vitamins A, D, E and K.

2. Why Lipids are a Better Source of Energy

Lipids contain a much higher proportion of carbon and hydrogen and very little oxygen compared with carbohydrates and proteins, so they can be oxidised more completely. On complete oxidation 1 g of lipid gives about 9 kcal (about 38 kJ) of energy, while 1 g of carbohydrate or protein gives only about 4 kcal. Lipids are also stored in the body in nearly water-free form, so more energy is stored per unit weight, making them the better energy source.

3. Saponification

Saponification is the hydrolysis of a fat or oil with an alkali to form glycerol and soap, which is the sodium or potassium salt of a long-chain fatty acid: triglyceride (fat/oil) + 3NaOH -> (heat) glycerol + 3RCOONa (soap). For example, glyceryl stearate boiled with NaOH gives glycerol and sodium stearate, a common soap.

4. Rancidity

Rancidity is the spoiling of fats and oils on standing in air, in which they develop a bad taste and unpleasant smell and become unfit to eat. It involves two simultaneous reactions: oxidation of the unsaturated fatty acid parts by atmospheric oxygen, and hydrolysis of the ester bonds by moisture, helped by enzymes and micro-organisms. The products include volatile short-chain fatty acids and aldehydes that cause the bad odour.

Minerals of Biological Significance

LESSON 60

Minerals are inorganic elements needed in small amounts for building body structures and for regulating enzymes, fluids, nerves and energy processes, and their deficiency causes specific diseases.

- **Calcium:** builds bones and teeth, helps blood clotting and muscle contraction; its deficiency causes weak bones.
- **Iron:** is the central part of haemoglobin and carries oxygen in blood; its deficiency causes anaemia.
- **Zinc:** is essential for the normal development and working of immune cells, for enzyme activity, growth and wound healing; its deficiency weakens immunity and makes a person more prone to infections.
- **Phosphorus:** with calcium, forms bones and teeth, and as part of ATP it stores and transfers energy, besides being a component of DNA, RNA and cell membranes.

Industrial Chemistry

UNIT II

Introduction to the Chemical Industry

LESSON 61

Industrial chemistry is the branch of chemistry which deals with the conversion of raw materials into useful products through chemical processes. Its primary objective is to investigate new ways and technologies to enhance the efficiency of chemical processes, reduce expenses and increase product yields. An industrial chemist plays a crucial role in enhancing productivity through the exploration of novel catalysts and optimising reaction conditions, while also aiming to reduce production costs by utilising economical raw materials. Industrial chemists also ensure safety standards, risk assessment and proper storage protocols. This multi-disciplinary field intersects with other scientific branches, including agriculture, engineering and pharmaceuticals.

1. Chemical Industries

The term chemical industries refers to all those companies that manufacture chemicals. These industries involve a broad range of activities including production of chemicals, quality assurance, packaging and distribution. Several chemical industries are involved in the manufacturing of basic chemicals, including acids, alkalis, salts, gases and more. Chemicals play an important role in the production of a wide range of products such as dyes, cosmetics, synthetic polymers, pesticides, cement, detergents, fertilisers, fibres, glasses, agrochemicals, etc. These industries encompass a wide range of sectors, generate substantial revenue and create employment opportunities. In Pakistan, the control and regulation of chemical industries involves government agencies such as the Pakistan Standards and Quality Control Authority (PSQCA) and the Environmental Protection Agency (EPA).

2. Main Types of Chemical Industries

- **Petrochemical industries:** primarily focused on the extraction, refining and processing of petroleum and natural gases. Crude oil on refining is used as fuel for transportation and power supply, and petrochemical feedstock or basic petrochemicals (methane, ethane, ethene, propene, butene, benzene, etc.) are converted into alcohols, carboxylic acids, amines, ethers, aldehydes and halogen derivatives, and then into final products such as solvents, polymers, fertilisers, pesticides, detergents, dyes and cosmetics, which meet the basic needs of clothing, shelter, food, health, transportation and communication. They also produce many other chemicals as by-products, which are then sold as raw materials for several useful substances.
- **Fertiliser industries:** responsible for the production of fertilisers, substances used to improve soil fertility and promote plant growth. Fertilisers provide essential nutrients to plants, enhancing crop yields and agricultural productivity. There are two main types: organic fertilisers, derived from natural sources, and synthetic fertilisers, manufactured chemically.
- **Textile industries:** sectors that involve the production of fabrics and textiles, including natural fibres like cotton and wool as well as synthetic materials like polyester and nylon, providing products from clothing and household items to industrial materials.
- **Paint industries:** produce liquid or powdered substances used for coating surfaces, providing protection, decoration and other functional purposes in construction, automotive and industrial sectors.

3. Synthetic Fertilisers and their Uses

Synthetic Fertilizer	Chemical Formula	Nutrients Provided	Uses
Ammonium Nitrate	NH_4NO_3	Nitrogen	Development of roots and maintaining pH level
Urea	$CO(NH_2)_2$	Nitrogen	Development of leaf, steam and fruits
Diammonium Phosphate (DAP)	$(NH_4)_2HPO_4$	Nitrogen, Phosphorus	Development of early stage of plants growth
Potassium Chloride (MOP)	KCl	Potassium	Help enzyme activities and photosynthesis

4. Synthetic Textile Products

Synthetic Textile Product	Description	Uses
Polyester	A synthetic fiber made from petrochemicals	Clothing (shirts, pants, dresses, jackets), curtains, bedding etc.
Nylon	A strong and durable synthetic fiber	Stockings, hosiery, swimwears, ropes, parachutes, and various sportswears
Acrylic	A soft and light weight synthetic fiber	Sweaters, blankets, faux fur, carpets

Pharmaceutical Industry

LESSON 62

1. Scope of the Pharmaceutical Industry in Pakistan

The pharmaceutical industry of Pakistan manufactures a large share of the medicines used in the country, including tablets, syrups, injections, antibiotics, painkillers and life-saving drugs, and it also exports medicines to many countries. It provides employment to a large number of chemists, pharmacists and workers and saves foreign exchange by reducing imports of finished drugs. Its scope is growing with increasing population and health needs, although it still depends largely on imported raw materials (active ingredients).

2. Important Drug Types

- **Antimalarial drugs:** chloroquine is the well-known drug used for the prevention and treatment of malaria caused by the Plasmodium parasite.
- **Analgesics and antipyretics:** aspirin is acetylsalicylic acid, prepared by acetylating salicylic acid with acetic anhydride; it acts as an analgesic (pain reliever), antipyretic (fever reducer) and anti-inflammatory drug. (Ascorbic acid is vitamin C.) Drugs that bring a raised body temperature (fever) back to normal are called antipyretics, common examples being paracetamol and aspirin.
- **Antihistamines:** an antihistamine is a drug that blocks the action of histamine, a chemical released in the body during allergic reactions. It is used for symptoms such as sneezing, running nose, itching, skin rashes and hives (urticaria), watery eyes and hay fever. Common examples are diphenhydramine and cetirizine; some older antihistamines also cause drowsiness and are used in cough and cold medicines. Ibuprofen and paracetamol are pain relievers and fever reducers, and diphenhydramine is an antihistamine used for allergy.

Pesticides

LESSON 63

Pesticides are chemicals used to kill or control pests that damage crops, stored food and health. Pesticides increase crop yield but must be used carefully because many are poisonous and persist in the environment.

1. Types of Pesticides

- **Insecticides:** kill insects, e.g. DDT (dichlorodiphenyltrichloroethane), widely used commercially to kill insects, especially mosquitoes spreading malaria.
- **Herbicides (weedicides):** kill unwanted weeds in fields, e.g. 2,4-D.
- **Fungicides:** kill fungi attacking crops, e.g. Bordeaux mixture.
- **Rodenticides:** kill rats and mice damaging stored grain.

Synthetic Polymers (PVC and Nylon)

LESSON 64

1. Natural and Synthetic Polymers

Two synthetic polymers are polyethene (used for bags and bottles) and nylon (used for fibres and ropes), both manufactured in industry from petrochemical monomers. Two natural polymers are cellulose (the structural material of plant cell walls, in wood and cotton) and starch (the energy-storage polymer of plants); cellulose is a natural polymer obtained from plants and is not a synthetic plastic, whereas nylon, Teflon and polyethene are synthetic polymers. Natural rubber and proteins are other examples of natural polymers.

2. Nylon 6,6

Nylon 6,6 is prepared by the condensation polymerisation of hexamethylene diamine (6 carbons) with adipic acid (6 carbons), with loss of water molecules and formation of amide links; the two 6-carbon monomers give it the name nylon 6,6. Nylon is a strong, elastic, abrasion-resistant polyamide fibre used for ropes, fabrics, carpets, brushes and machine parts.

3. Polyvinyl Chloride (PVC)

PVC is prepared by addition polymerisation of vinyl chloride CH2=CHCl: nCH2=CHCl -> -[CH2-CHCl]n-. PVC is hard, tough, chemically resistant, non-flammable and a good electrical insulator; it is used for water and drainage pipes, electric cable insulation, floor tiles, raincoats, bottles and plastic sheets.

4. Thermosetting Plastics

Bakelite, made from phenol and formaldehyde, is an example of a thermosetting plastic: it forms a hard cross-linked network on heating and cannot be softened or remoulded again. Polyethene, PVC and nylon are thermoplastics that soften on heating and can be remoulded.

Cosmetics: Lipsticks, Nail Polish and Remover, Perfumes

LESSON 65

Cosmetics are substances applied to the body, especially the face, skin, hair and nails, to clean, protect, beautify and improve appearance.

1. Commonly Used Cosmetics

- **Nail polish:** a coloured lacquer of nitrocellulose, resin, plasticiser and solvent applied on nails. Its four main components are: (1) a film-forming polymer, usually nitrocellulose, which forms the hard glossy film on the nail; (2) a resin, which improves adhesion and gloss; (3) a plasticiser such as camphor or dibutyl phthalate, which keeps the film flexible and prevents cracking; and (4) solvents such as ethyl acetate or butyl acetate, plus pigments and colours that give the polish its shade.
- **Nail polish remover:** consists mainly of acetone, a powerful solvent that quickly dissolves the film-forming polymer and pigments of dried nail polish.
- **Lipstick:** a mixture of waxes, oils and pigments applied to the lips.
- **Perfumes:** solutions of fragrant oils in alcohol that give a pleasant smell.
- **Hair dyes:** colour the hair; permanent dyes use hydrogen peroxide H2O2 as the oxidising agent (developer), which oxidises the dye precursors so that large coloured molecules form inside the hair, and also lightens the natural pigment. Resorcinol is a dye coupler used in them, not the oxidiser.

Other common cosmetics include face creams, lotions and powders.

Adhesives

LESSON 66

Adhesives are substances used to join two surfaces together by bonding, for example glues, gums and resins used in woodwork, packaging and industry.

1. Super Glue

Super glue is chemically based on cyanoacrylate (methyl or ethyl cyanoacrylate), which polymerises within seconds on contact with the moisture present on surfaces. It therefore forms an extremely strong bond between almost any materials, even skin, almost instantly, so it is used for quick repairs and even in surgery, unlike the slower-setting epoxy resins or polyvinyl acetate adhesives.

2. Silicone Resins

Silicone resins (the book writes silicon resins) are heat-resistant, water-repellent adhesives and sealants that remain flexible over a wide temperature range, used for sealing joints, in electronics and in high-temperature applications.

Environmental Chemistry

UNIT 12

Chemistry of the Troposphere

LESSON 67

The atmosphere is divided into layers: the troposphere extends from the surface to about 11 km, the stratosphere from 11 km to 50 km, the mesosphere from 50 to 85 km and the thermosphere from 85 to 500 km. We live in the lowest layer, the troposphere, which contains the air we breathe and where weather phenomena like rain, clouds and storms occur.

1. Chemistry of Nitrogen and Sulphur Oxides in the Troposphere

Nitrogen and its oxides react as: N2 + O2 -> (lightning/engine heat) 2NO, and 2NO + O2 -> 2NO2; NO2 with water and sunlight forms nitric acid and ozone, contributing to acid rain and photochemical smog, e.g. 4NO2 + 2H2O + O2 -> 4HNO3. Sulphur in coal and oil burns to SO2: S + O2 -> SO2, which oxidises further: 2SO2 + O2 -> 2SO3, and SO3 + H2O -> H2SO4 (sulphuric acid), or SO2 + H2O -> H2SO3. These acids return as acid rain, while NOx with hydrocarbons and ozone together produce smog that harms lungs, crops and materials.

2. Smog

The word smog is made from smoke + fog. Industrial smog is the grey, harmful smog found over industrial cities, formed mainly from the burning of coal and fossil fuels. It is a mixture of fog, smoke particles, ash and sulphur dioxide; the SO2 oxidises and combines with water droplets to form sulphuric acid aerosol: 2SO2 + O2 -> 2SO3, SO3 + H2O -> H2SO4. It forms especially in cold, damp, still weather under temperature inversion, reduces visibility and causes respiratory diseases and damage to buildings.

3. Acid Rain

Normal rain is slightly acidic (pH about 5.6) because of dissolved CO2, but acid rain contains sulphuric acid and nitric acid formed from SOx and NOx pollutants, which lowers its pH further, typically below 5 and often near 4. Acid rain harms human health mainly indirectly: it dissolves toxic metals like aluminium, lead and mercury from soil into drinking water, irritates the eyes, skin and respiratory system through its precursor gases SO2 and NOx, and damages crops and water supplies on which people depend. It also corrodes buildings, monuments and metal structures. Prevention measures include reducing SO2 and NOx emissions by using low-sulphur fuels, fitting scrubbers in factory chimneys and catalytic converters in vehicles, shifting to cleaner and renewable energy sources, and enforcing environmental laws through agencies like the EPA.

4. Greenhouse Effect and Global Warming

The greenhouse effect is the warming of the Earth's atmosphere by gases that allow sunlight to reach the surface but absorb the infrared (heat) radiation re-emitted by the Earth, like the glass of a greenhouse; in natural amounts it keeps the Earth warm enough for life. The main greenhouse gases are carbon dioxide CO2 (from burning fossil fuels and deforestation), methane CH4 (from decaying matter, livestock and rice fields), nitrous oxide N2O (from fertilisers), water vapours, ozone O3 and chlorofluorocarbons CFCs. Oxygen, nitrogen and argon, the main components of air, do not absorb infrared radiation in this way. Human activities enhance the effect: burning fossil fuels in industries, vehicles and power plants releases huge amounts of CO2, deforestation reduces CO2 absorption, agriculture and livestock release methane and nitrous oxide, and industries release CFCs. The main cause of the rise in global average temperature (global warming) is this increased emission of greenhouse gases, chiefly carbon dioxide. The trapped heat disturbs weather patterns: heatwaves become more frequent, rainfall becomes irregular with more floods and droughts, glaciers and polar ice melt raising sea level, and storms and extreme weather events become stronger and less predictable.

5. Catalytic Converters

A catalytic converter is a device fitted in the exhaust system of automobiles, containing a honeycomb coated with platinum, palladium and rhodium catalysts (mainly Pt and Pd), which converts harmful exhaust gases into harmless ones at high temperature. The main pollutants it targets are carbon monoxide CO (converted to CO2: 2CO + O2 -> 2CO2), unburnt hydrocarbons (converted to CO2 and H2O) and nitrogen oxides NOx (converted to N2 and O2: 2NO -> N2 + O2). It requires unleaded petrol, because lead poisons the catalyst.

Chemistry of the Stratosphere

LESSON 68

The stratosphere extends from 11 km to about 50 km above the Earth. The ozone layer lies in the stratosphere, about 15 to 35 km above the Earth, where ozone O3 is formed from oxygen by ultraviolet radiation. This layer absorbs most of the harmful UV radiation of the Sun and protects life on Earth from it.

1. Causes of Ozone Depletion

Ozone depletion in the upper atmosphere is mainly caused by chlorofluorocarbons (CFCs). The main cause is the release of ozone-depleting substances, especially CFCs from refrigerators, air conditioners, aerosol sprays and foam making, and also halons and nitrogen oxides. In the stratosphere, UV radiation breaks CFCs and releases chlorine atoms: CFCl3 -> (UV) CFCl2 + Cl. The chlorine atom destroys ozone in a repeating chain: Cl + O3 -> ClO + O2 and ClO + O -> Cl + O2, so one chlorine atom can destroy about 100,000 ozone molecules, thinning the ozone layer.

2. Ozone-Depleting Substances (ODS)

Ozone-depleting substances are man-made chemicals that release chlorine or bromine in the stratosphere and destroy ozone, mainly chlorofluorocarbons (CFCs), halons, carbon tetrachloride, methyl chloroform and HCFCs. Human activities that contributed include the wide use of CFCs as refrigerants in refrigerators and air conditioners, as propellants in aerosol sprays and in foam blowing, halons in fire extinguishers, and nitrogen oxides from jet aircraft and fertilisers. Because of the resulting ozone hole and increased UV radiation, the Montreal Protocol has phased out most ODS and replaced them with safer alternatives like HFCs and hydrocarbons.

Water Pollution and Water Analysis

LESSON 69

1. Sources of Water Pollution

The major source of water pollution is industrial activities: industries discharge untreated effluents containing toxic chemicals, heavy metals, dyes, acids and oils into rivers and groundwater. Domestic sewage and agricultural run-off of fertilisers and pesticides are the other major sources.

2. Testing of Waste Water

- **pH measurement:** shows whether the water is acidic or alkaline compared with safe limits.
- **BOD (biochemical oxygen demand):** the oxygen used by micro-organisms to decompose organic matter in a fixed time; higher BOD means more organic pollution.
- **COD (chemical oxygen demand):** the oxygen needed to chemically oxidise the pollutants with a strong oxidising agent.
- **Dissolved solids and specific pollutants:** determination of dissolved solids and pollutants such as heavy metals, oil and grease, so the water can be compared with WHO and environmental standards before discharge.

Green Chemistry

LESSON 70

The primary goal of green chemistry is to design chemical products and processes that are safe for human health and the environment, preventing pollution at its source - that is, developing environmentally friendly chemicals. Its goals include developing environmentally friendly chemicals, minimising waste generation and hazards, reducing energy consumption, and using renewable raw materials and safer solvents. Green chemistry is applied in industry together with measures such as fitting catalytic converters in vehicles, using low-sulphur fuels, scrubbing factory chimney gases and shifting to cleaner and renewable energy sources, so that harmful emissions of SOx, NOx, CO and unburnt hydrocarbons are prevented rather than treated afterwards.

Spectroscopy

UNIT 13

Methods of Spectroscopy

LESSON 71

1. The Electromagnetic Spectrum

In the electromagnetic spectrum, energy increases in the order radio waves, microwaves, infrared, visible, UV, X-rays, gamma rays ($E = h\nu$, so higher frequency means higher energy). IR radiation therefore has energy higher than radio waves and microwaves, but lower than UV waves and X-rays. Each region causes a different change in a molecule: UV-visible radiations have enough energy to raise electrons from a lower energy level to a higher one (electronic excitation or electronic transition); IR radiation only increases bond vibrations; and radio waves only change nuclear spin.

2. UV-Visible Spectroscopy

The purpose of UV-visible spectroscopy is to study molecules that absorb UV or visible light through electronic transitions, and thereby to identify conjugated systems and measure the concentration of a substance from its absorbance. The transitions, in decreasing order of energy required, are: sigma to sigma* (highest energy, because sigma bonds are very strong and their electrons are held tightly, absorbing at the shortest wavelength in the far UV), then n to sigma*, then pi to pi*, and n to pi* (lowest). In chemistry the technique is used for structure determination, studying reaction kinetics, detecting impurities and in quantitative analysis and quality control of drugs. In biology it is used to measure proteins and nucleic acids (DNA absorbs at 260 nm and proteins at 280 nm), to study enzymes and to estimate the concentration of biological samples.

Lambda max: lambda max is the wavelength at which a substance shows the maximum absorption of UV-visible radiation in its spectrum; it is a characteristic property of the compound and the transition involved. In the UV-visible spectrum of ethanol shown in the book (absorbance plotted against wavelength in nm, 200-500 nm), the curve rises to a single strong peak with maximum absorbance of about 1.5 at about 270-280 nm and then falls to nearly zero above 300 nm. The wavelength at the top of this peak is the lambda max of ethanol, arising from the n to sigma* transition of the lone-pair electrons of the -OH oxygen, and ethanol shows no significant absorption in the visible region.

3. Infrared (IR) Spectroscopy

IR spectroscopy is the technique used to determine the functional groups in a given organic molecule: each functional group absorbs IR radiation at a characteristic wavenumber (for example C=O at about 1680-1740 cm-1 and O-H at about 3600-3650 cm-1). By noting which absorptions are present, IR spectroscopy identifies the bonds and functional groups present. In the IR spectrum frequency is expressed as wavenumber, the number of waves per centimetre, whose unit is reciprocal centimetre, cm-1; the useful IR range is about 4000-670 cm-1, while nanometre and angstrom are units of wavelength used in UV-visible spectroscopy. The fingerprint region, typically 1500-500 cm-1, contains many bending and skeletal vibrations and gives a complex pattern that is unique for each molecule, like a fingerprint of a person; it is used to identify a compound by comparing its spectrum with that of a known sample.

4. Proton NMR Spectroscopy

Proton NMR (1H NMR) spectroscopy is the study of the absorption of radio waves by hydrogen nuclei of a molecule placed in a strong magnetic field. It detects nuclei that have spin, principally the hydrogen nucleus 1H (a single proton) present in the sample. Only nuclei with an odd number of protons or neutrons, such as 1H and 13C, have a magnetic spin and can absorb radio waves in a magnetic field and be detected. The protons act like tiny magnets and align with or against the field; when they absorb radiofrequency energy of the right frequency they flip to the higher energy state (resonance). Protons in different environments absorb at slightly different frequencies (chemical shift, in ppm, with TMS as reference at 0 ppm), and the number of signals, their splitting and their areas reveal the structure of the molecule. In a proton NMR spectrum the x-axis represents the chemical shift, delta, in ppm, increasing from right (0 ppm, the TMS reference) to left (downfield, up to about 10-12 ppm), and the y-axis represents the intensity or strength of absorption (signal height/area), which is proportional to the number of equivalent protons producing that signal. D2O (heavy water) is used as a solvent in NMR spectroscopy because deuterium does not give a proton NMR signal, so it does not hide the signals of the sample; it is also used to identify -OH and -NH protons, which exchange with deuterium and their peaks disappear from the spectrum. NMR is applied in determining the structure of organic compounds, identifying ethanol and other substances, studying proteins, medical imaging (MRI) and quality control.

NMR spectrum of ethanol: the proton NMR graph of ethanol shows three signals because ethanol has three kinds of protons: CH3, CH2 and OH, in the ratio 3:2:1. Peak a at about 1.2 ppm is the CH3 group, split into a triplet by the two neighbouring CH2 protons, with the largest area (3 protons). Peak c at about 3.7 ppm is the CH2 group, split into a quartet by the three CH3 protons, with area for 2 protons; it is further downfield because it is directly attached to oxygen. Peak b at about 4.6 ppm is the single OH proton, appearing as a broad singlet with the smallest area (1 proton), most deshielded because it is bonded to electronegative oxygen. In the book, the hydroxyl proton of C2H5OH appears as a broad singlet around 4-5 ppm; its exact position varies with concentration and solvent because of exchange of the OH proton. Together the positions, splitting and areas of a, b and c confirm the structure CH3-CH2-OH.

5. Mass Spectroscopy

Mass spectroscopy gives the accurate relative molecular mass of a compound from its molecular ion peak (M+), from which its molecular formula can be worked out. The fragmentation pattern - the smaller peaks at different mass-to-charge (m/z) values - shows how the molecule breaks up, which reveals the groups and arrangement of atoms present and can distinguish between isomers. Its applications include identification of unknown compounds, structure determination, purity checking, detection of drugs and forensic materials, isotope abundance measurement, and quality control in the pharmaceutical and food industries.

6. Atomic Absorption and Emission Spectroscopy

In atomic emission spectroscopy the sample is heated or excited so that its atoms emit light of characteristic wavelengths on returning to lower energy levels, giving bright lines on a dark background. In atomic absorption spectroscopy light from a lamp is passed through the vapour of the sample and the atoms absorb their characteristic wavelengths, giving dark lines against a bright background. Emission measures the light given out and is used mostly for qualitative detection, while absorption measures the light absorbed and is widely used for quantitative determination of metals.